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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)	Prepared By		Shreyash K.	22.07.2026
	Verified By		Prafulla Sarode	22.07.2026

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DRAFT ACCEPTANCE TEST PROCEDURE
FOR
INTELLIGENT VEHICLE INTERCOM SYSTEM - iVIS
(UCZN10331)

DS Cat. No. _____

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Prepared & published by

ELCOM Innovations Pvt. Ltd.
B-76, Industrial Area, Phase VII,
SAS Nagar, Punjab-160 055 (India)

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PA2	Alignment and formatting of Document	25/07/2025	MCJ	Prafulla Sarode
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List of Abbreviations

AATV	Armored Ambulance Tracked Vehicle
ALNS	Automatic Land Navigation System
ANT	Antenna
ATP	Acceptance Test Procedure
BITE	Built-In Test Equipment
CBL	Cable
CDR / CMDR	Commander
CH	Channel
CMTV	Carrier Mortar Tracked Vehicle
CO-DVR	Co-Driver
COM	Communication Port
CSU	Central Switch Unit
DC	Direct Current
DCH	Digital Control Harness
DEMO	Demonstration
DOC	Doctor
DVR	Driver
ETH	Ethernet
GPS	Global Positioning System
IO	Infantry Operator
IP	Internet Protocol
iVIS	Intelligent Vehicle Intercom System
KBRD	Keyboard
MGMT	Management
OLED	Organic Light Emitting Diode
OS	Operating System
PC	Personal Computer
PSU	Power Supply Unit
PTT	Push-to-Talk
PWR	Power
Qty.	Quantity
QWERTY	Standard keyboard layout named after the first six keys
R/A	Radio/Aux Connector
RAW	Raw audio/data file format
RJ45	Registered Jack 45 Ethernet connector
RS232	Recommended Standard 232 serial communication interface
RTP	Real-time Transport Protocol
S. No.	Serial Number
SPE	Single Pair Ethernet
SSH	Secure Shell
TMR	Test Measurement Report
UDP	User Datagram Protocol
UHF	Ultra-High Frequency
USB	Universal Serial Bus
UU	User Unit
VDC	Volts Direct Current
VL	Volume Level

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1 Scope & Introduction

1.1 Purpose

The purpose of this document is to present a test procedure for acceptance of all the components of a Digital Control Harness (Intercom System - iVIS) for an Armored Vehicle. This Digital Control Harness is referred to as iVIS with model number as UCZN10331.

1.2 Introduction

iVIS (Intelligent Vehicle Intercom System) is a family of IP-based Vehicle Intercom Systems designed for deployment across both Tracked and Wheeled Vehicles. The iVIS is lightweight, compact and provide seamless communication among vehicle crew members. The system is designed for simplicity and ease of operation.

The system interfaces with in-service vehicle radios, allowing crew members to communicate externally with Command & Control Center. Its IP-based architecture ensures enhanced flexibility, scalability, and reliable communication in demanding operational environments. The iVIS is engineered to operate reliably under harsh environmental conditions and sustains a wide operating temperature range from -20°C to +55°C.

The system incorporates polarity reversal protection and is equipped with transient suppression mechanisms to safeguard against voltage spikes, high surges, and other electrical disturbances, thereby ensuring dependable performance and protection of critical electronics in rugged vehicular applications.

1.3 Scope

The scope of this document is to define various tests to be conducted on the **iVIS** to verify its compliance with the specified functional and performance requirements. The system provides all communication capabilities expected from an Armored Vehicle Intercom System, enabling reliable internal and external communications within operational environments.

Upon completion of the testing activities, all observations, measurements, and results shall be documented and recorded in the **Test Measurement Report (TMR)**.

This document covers all tests required for the acceptance and validation of the iVIS system as a Vehicle Intercom System. The Acceptance Test Procedure (ATP) includes, but is not limited to, the following test categories:

- Physical Inspection
- Electrical Checks
- Functional Tests

Detailed test procedures, instructions for conducting the tests, acceptance criteria, and expected results are provided in the subsequent sections of this document.

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2 System Description

2.1 Brief Description of the Equipment

iVIS is an intelligent Vehicle Intercom System customized for armored vehicles. iVIS provides a facility for various crew members of vehicles to communicate with each other. It also provides a facility to communicate over the Radio system as well.

2.2 Components of iVIS

iVIS intercom system consists of following sub-components:

- 1 Central Switch Unit
- 2 User Units
- 3 Set of cables (Connecting and Power cable)
- 4 iVIS Keyboard Unit (Optional External QWERTY Keyboard)
- 5 Stick Radio with Docking Station

2.3 iVIS operation details

iVIS has the following operational endpoints/users.

- 1 User Headsets one each for each user unit
- 2 QWERTY Keyboard
- 3 Radio Interface for Vehicle's Radio System – 2 no.

For voice communication, these interfaces are programmed in such a way that intercom as well as interacting with Radio systems is very easy.

2.4 Operation Modes

iVIS system has the following modes of operation including 1 Intercom Mode and multiple Radio Modes

1. **Intercom Mode** – In this mode all User/Crew members can talk with each other in conference mode of voice communication.
2. **Radio Modes** – There are total of four radios connections. If access is granted for Radio access to particular User/Crew, he can select desired radio and communicate with the radio system.
3. **Radio listen facility** – By way of this facility, User/Crew members can listen to Vehicle's Radio system even when they are in Intercom Mode. By default, this facility is provided to all User/Crew headsets.

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3 Safety & Security

3.1 Safety and Security Aspects

The DCH has been designed with due consideration of operational safety, reliability, and information security requirements applicable to vehicle-based communication systems.

Safety Aspects

- The DCH operates within the specified input voltage range and incorporates protection against reverse polarity, over-voltage, under-voltage, and transient surges.
- Internal protection mechanisms safeguard the equipment against electrical faults, thereby minimizing the risk of damage to connected subsystems.
- The unit is designed to withstand harsh environmental conditions, including vibration, shock, temperature variations, and humidity encountered in vehicular applications.
- All connectors and interfaces are mechanically secured to prevent accidental disconnection during vehicle movement and operation.
- The compact and ruggedized construction ensures safe installation and operation within the vehicle platform.

Security Aspects

- The DCH supports secure communication between connected subsystems and prevents unauthorized access to network resources.
- Access to configuration and maintenance functions is restricted to authorized personnel only.
- The system architecture minimizes the possibility of unauthorized modification of operational parameters.
- Communication interfaces are designed to prevent unintended data leakage between connected networks.
- System logs and event records, where applicable, can be maintained for monitoring and diagnostics purposes.

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4 Technical Description of iVIS Component

This chapter describes various sub-components of iVIS Intercom System. This chapter will elaborate about physical structure, various interfaces available, and operational details of each of the sub-components of iVIS.

4.1 iVIS Central Switch Unit

This is the Central switch unit where the power is terminated. This unit distributes power as well as Ethernet connections to other units required for establishing communication. This unit also has 4 radio interface facilities to interface with Vehicle Radio System for communication over radio link. For management and maintenance purposes, a laptop/PC can be connected to the system through a central switch unit via RJ45 MGMT connector interface.

The following is the image of the Central Switch Unit.

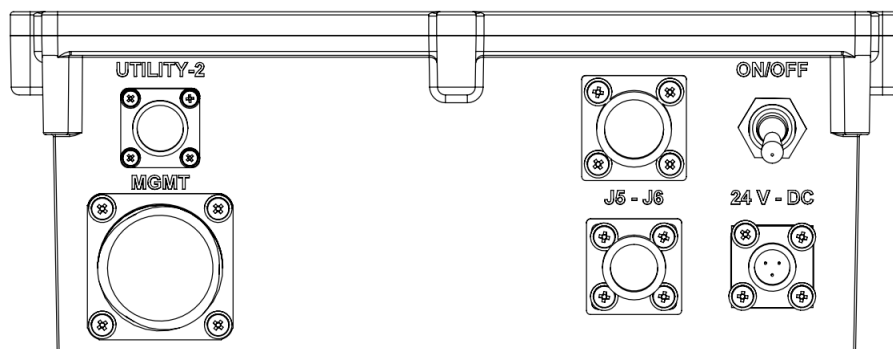


Figure 1: iVIS Central Switch Unit Top View

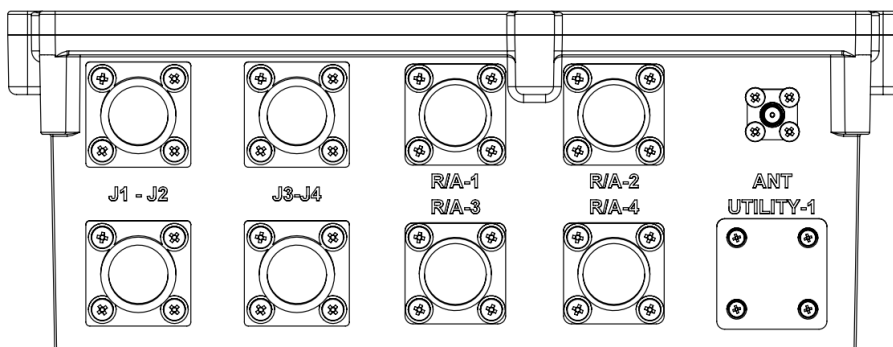


Figure 2: iVIS Central Switch Unit Bottom View

The Radio interface present in the Central Switch Unit enables all users to communicate over Radio link.

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Physical parameters

The following figure shows various physical parameters of the Central Switch Unit.

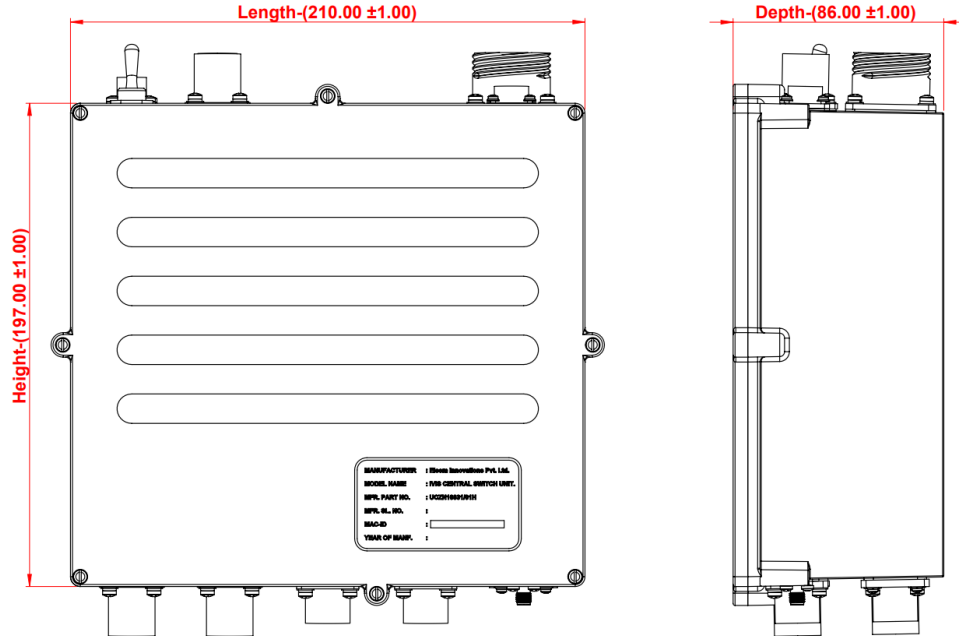


Figure 3: Dimension – iVIS Central Switch Unit

The following table lists various physical parameters of the Central Switch Unit

S. No	Dimension Description	Measure and tolerance	Location
1	Length	210 mm (± 1 mm)	-
2	Height	197 mm (± 1 mm)	-
3	Depth	86 mm (± 1 mm)	-
4	Weight	2.250 ± 0.200 Kg	-
5	Colour	Olive-Green	-
6	Radio Connector	4 nos (R/A-1 to R/A4)	Bottom Side
7	ETH & Power Connector	4 nos (J1 to J4)	Bottom Side
		1 no (J5)	Top Side
8	SPE & Power Connector	1 no (J6)	Bottom Side
9	Input Power Connector	1 no (24V-DC)	Top Side
10	Power On/Off Switch	1 no (ON/OFF)	Top Side
11	MGMT ETH Connector	1 no (MGMT)	Top Side
12	Utility-1	0 no (UTILITY-1)	Bottom Side
13	Utility-2	1 no (UTILITY-2)	Top Side
14	Antenna Connector	1 no (ANT)	Bottom Side

Table 1: Physical Parameters – iVIS Central Switch Unit

Note: The result record in TMR Sheet

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Note - For ease of understanding, dimensions are taken considering unit is wall mounted & all connectors are Up or down facing.

4.2 iVIS User Unit

This is a user interface unit to be installed in front of the user inside the vehicle. User Unit can work as Commander Unit, Driver Unit or any other User Unit. This unit is connected to the Central Switch unit using the Data-Power Cable. This unit gets power from the Central Switch Unit.

Basic function of User Unit is to provide headset connection to User's Headset and allow user to talk on intercom group or select radio system access to talk on Radio system etc. User Unit has volume knob to control the self-headset speaker volume and channel selection knob using which user can select operation between Intercom or Radio. The selected knob position also decides whose voice user can listen to on. User Unit has OLED display to present various information to User. User Unit also has interface to connect to optional iVIS QWERTY Keyboard Unit.

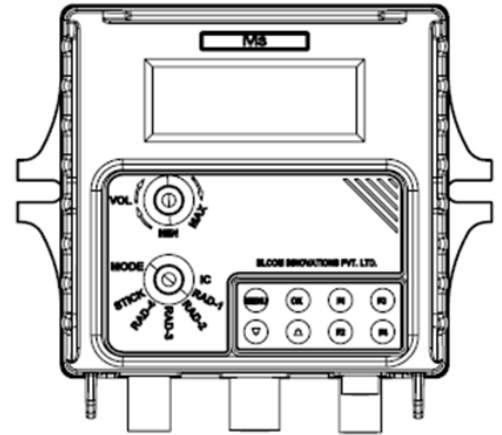


Figure 4: iVIS User Unit

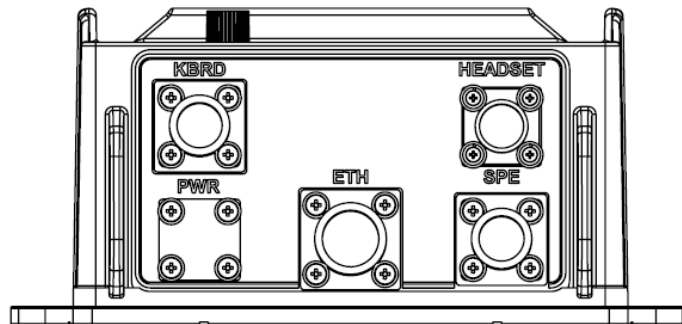


Figure 5: iVIS User Unit Connectors view

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Physical parameters of User Unit

The following table lists various physical parameters of the User Unit.

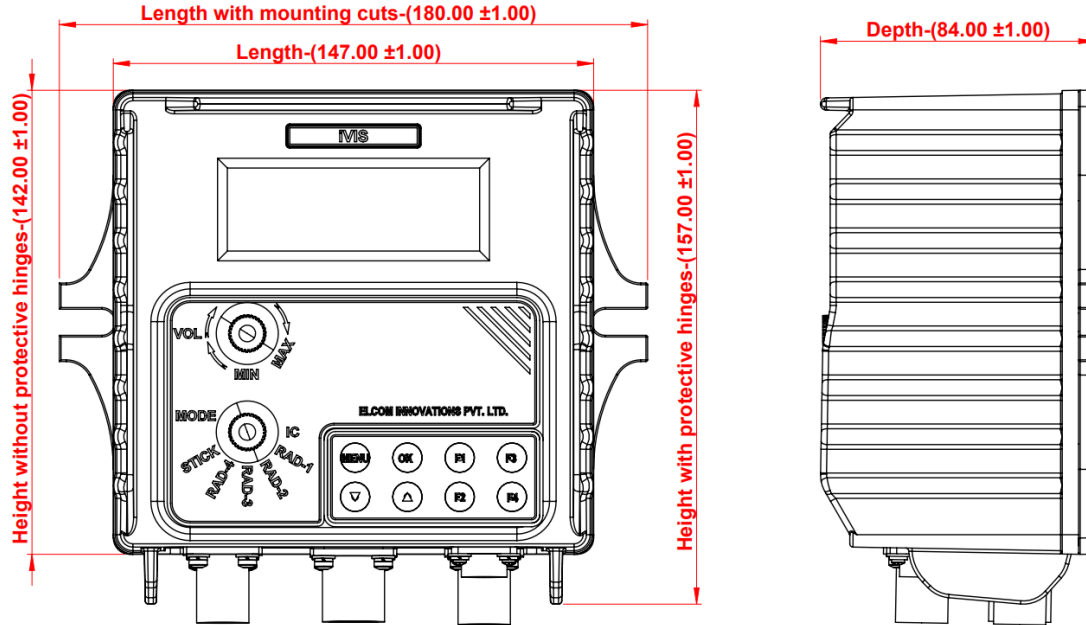


Figure 6: Dimension – iVIS User Unit

The following table lists various physical parameters of the User Unit.

S. No	Dimension Description	Measure and tolerance	Location
1	Length	147 ± 1 mm	-
2	Length with mounting cuts	180 ± 1 mm	-
3	Height without protective hinges	142 ± 1 mm	-
4	Height with protective hinges	157 ± 1 mm	-
5	Depth	84 ± 1 mm	-
6	Weight	1.500 ± 0.100 Kg	-
7	Colour	Olive-Green	-
8	Headset Connector	1 no (Headset)	Bottom Side
9	ETH & Power Connector	1 no (ETH)	Bottom Side
10	SPE & Power Connector	1 no (SPE)	Bottom Side
11	QWERTY Keyboard Connector	1 no (KBRD)	Bottom Side
12	Input Power Connector	0 no (PWR)	Bottom Side
13	Volume Knob	1 no	Front Side
14	Volume Knob Position	11 positions	Front Side
15	Mode Knob	1 no	Front Side
16	Mode Knob position	6 positions	Front Side
17	OLED Display	1 no	Front Side
18	Built-In Keypad Buttons	8 no's	Front Side

Table 2: Physical Parameters – iVIS User Unit

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Note: The result record in TMR Sheet

4.3 QWERTY Keyboard Unit

This QWERTY Keyboard Unit is to be connected to the User Unit. QWERTY Keyboard Unit comes with the connecting cable by which it can connect and can work with all the user units.

Basic function of QWERTY Keyboard Unit is to provide typing features to User and allow user to send text message. Keyboard also has multiplexed functions and multimedia keys in number row.

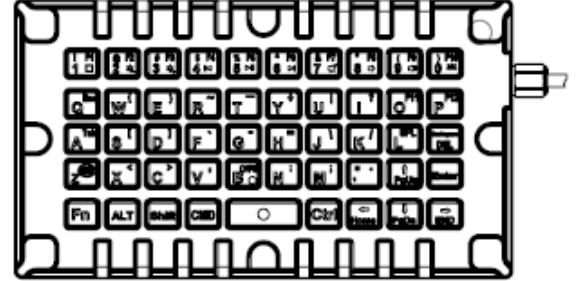


Figure 7: QWERTY Keyboard

QWERTY Keyboard Unit has backlit with auto-off function.

Physical parameters of QWERTY Keyboard Unit

The following figure shows physical parameters of the QWERTY Keyboard Unit.

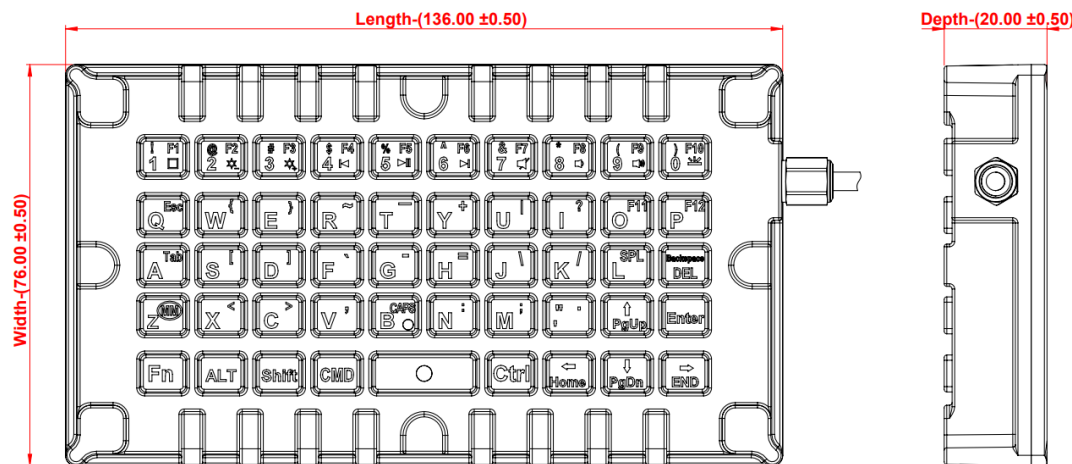


Figure 8: Dimension – QWERTY Keyboard

The following table lists various physical parameters of the QWERTY Keyboard Unit.

S. No	Dimension Description	Measure and tolerance	Location
1	Length	136.00 ± 0.50 mm	-
2	Width	76.00 ± 0.50 mm	-
3	Depth	20.00 ± 0.50 mm	-
4	Color	Olive-Green	-
5	Keyboard Buttons	49 Nos	Top Side
6	Weight	0.270 ± 0.050 kg	-

Table 3: Physical Parameters – QWERTY Keyboard Unit

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		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

Note: The result record in TMR Sheet

4.4 Stick Radio

iVIS supports a dedicated UHF Stick Radio to enable seamless communication between the vehicle crew and dismounted personnel operating outside the vehicle. The Stick Radio communicates through the vehicle-mounted antenna interface, providing reliable line-of-sight voice communication over distances exceeding 1500 meters, depending on terrain and operating conditions.

The Stick Radio is a compact handheld transceiver featuring programmable channels, Push-to-Talk (PTT) operation, adjustable volume control, channel selection, and status indication. It operates on rechargeable battery power and is stored in a dedicated Stick Radio Dock Station inside the vehicle, where it is securely held and charged whenever not in use. This arrangement ensures that the radio remains fully charged and readily available for rapid deployment during operational missions.

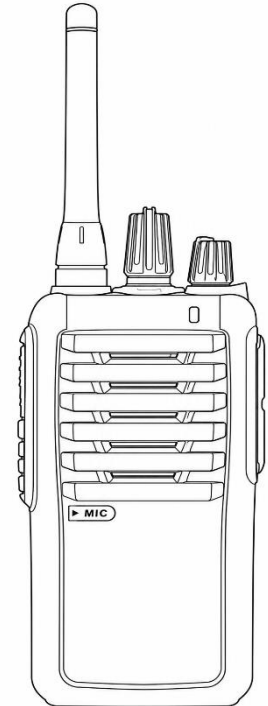


Figure 9: Stick Radio

4.5 Stick Radio Dock Station

The Stick Radio Dock Station serves as the charging and storage unit for Stick Radio. It receives power directly from the vehicle power supply and provides the required power for battery charging. The dock securely holds the radio during vehicle operation while ensuring the battery remains charged and ready for immediate use.

The Stick Radio Dock Station enables convenient charging without removing the battery from the radio and provides a dedicated location for safe storage when the Stick Radio is not in use. It is designed for reliable operation under vehicle operating conditions and ensures the radio is readily available for deployment.

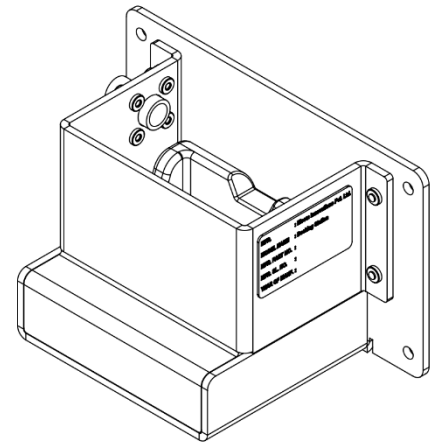


Figure 10: Stick Radio Dock Station

Physical Parameters of Stick Radio Dock Station

The following figure shows physical parameters of the Stick Radio Dock Station.

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

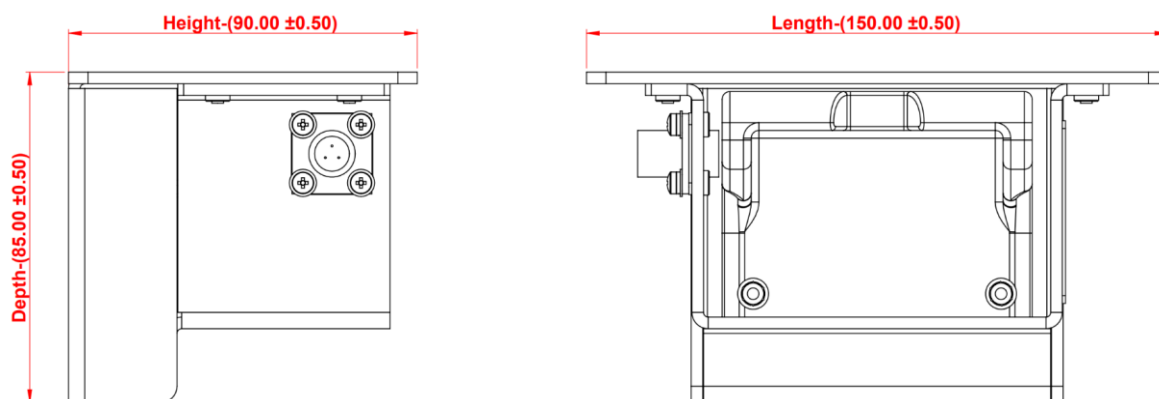


Figure 11: Stick Radio Dock Station – Dimension

The following table lists various physical parameters of the Stick Radio Dock Station.

S. No	Dimension Description	Measure and tolerance	Location
1	Length	150.00 ± 0.50 mm	-
2	Height	90.00 ± 0.50 mm	-
3	Depth	85.00 ± 0.50 mm	-
4	Color	Olive-Green	-
5	Input Power Connector	1 Nos	Left Side
6	Weight	0.270 ± 0.050 kg	-

Table 4: Stick Radio Dock Station – Physical Parameters

Note: The result record in TMR Sheet

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

5 Instruments and Accessories required for Testing

The test will be carried out with the delivered equipment with following additional measuring equipment and cable accessories.

5.1 List of Test & Measuring Equipment and Support Accessories

The test equipment listed in

Table 5 or their equivalent shall be used for measurement and monitoring.

S. No	Description	Part No / Model No	Qty
1	Laptop/ PC with Window 10 or above.	Any Make	1 No.
2	Multi meter	Any make	1 No.
3	Measuring tape-5m	Any make	1 No
4	Vernier calliper	Any make	1 No
5	Weighing Machine	Any Make	1 No
6	24 V DC power source	Any make	1 No
7	Radio Base station with PSU	Motorola GM338	1 No
8	Radio Handheld	Motorola	1 No
9	Charger for Radio Handheld	Motorola	1 No
10	USB to serial RS232 adapter	Any make	1 No
11	iVIS Central Switch Unit	Elcom Innovations Pvt Ltd	1 No
12	iVIS User Unit	Elcom Innovations Pvt Ltd	1 No

Table 5: List of Test & Measuring Equipment and Support Accessories

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

5.2 List of Cables for Testing

List of cables and accessories required are mentioned in Table 6.

S. No	Identification marking	Purpose	Qty.
1	CBL21007	Radio Cable – CSU to Mortorola	2
2	RJ45 Ethernet Cable	To connect PC/Laptop to the setup	1
3	SPE & Power cable Open Ended	To demonstrate Shot circuit Test	1
4	Utility Interface Test cable	To demonstrate (ALNS)/GPS Interface	1

Table 6: List of Test Cables

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		Verified By	Prafulla Sarode	22.07.2026

6 Test Sequence

Following table elaborate on the kind of tests to be conducted.

S. No	Test Name	Description
1	Physical Inspection and verification	Verification of physical dimensions, color, interfaces etc.
2	Power-on test/Electrical test	Power-On time and power consumption test
3	Version verification	All the equipment is subject to OS, firmware and software versions verification
4	Interfaces verification	In this test all the interfaces are tested. Interfaces are power connectors, Data-Power connectors, headset connectors, Volume Knob, Mode Knob, OLED display etc.
4.1	BITE tests	Verification of various BITE functions for system performance
4.1.1	Display Test	Perform Display BITE Test according to Para 8.7.2
4.1.2	Functional Keypad	Perform Functional Keypad BITE Test according to Para 8.7.3
4.1.3	PTT Test	Perform PTT BITE Test according to Para 8.7.4
4.1.4	Headset Mic Test	Perform Headset Mic BITE Test according to Para 8.7.6
4.1.5	Keyboard Test	Perform Keyboard BITE Test according to Para 8.8
4.2	Functional Test	Functional tests as per below listed sub tests
4.2.1	Volume Knob	Perform Volume Knob Test according to Para 8.7.1
4.2.2	Mode Knob	Perform Mode Knob Test according to Para 8.7.1
4.2.3	Speaker Test	Perform Headset Speaker Test according to Para 8.7.5

Table 7: Test Sequence and Sample size for tests

Note: -

- All parametric tests shall be conducted in the in-house laboratory using calibrated environmental test equipment, including temperature chambers and vibration test systems.
- Functional tests shall be conducted in the factory laboratory under ambient room temperature conditions with the test setup configured on a tabletop arrangement.
- The power source used for testing shall be a 24 VDC supply derived from an AC-DC converter rated at 100 Watts or higher.
- All measuring instruments and test equipment used during testing shall possess valid calibration certificates traceable to national or international standards.
- Any deviations from the test procedures or acceptance criteria specified in this document shall be recorded and documented in the Test Measurement Report (TMR).

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		Verified By	Prafulla Sarode	22.07.2026

7 Software

7.1 List of Software

S. No	Description	Version No.	Source
1	PuTTY	0.84 or later	Freeware
2	Wireshark	4.6.6 or later	Freeware
3	Audacity	3.7.8 or later	Freeware

Table 8: Application Software

7.2 PC set up/configuration for Connectivity and tests through login

1. Configure the PC/Laptop with IP address 192.168.2.101
2. Default IP addresses for various sub-units in CMTV
 - a. Central Switch Unit - IP address 192.168.2.150
 - b. User Unit 1 (CDR) - IP address 192.168.2.151
 - c. User Unit 2 (IO) - IP address 192.168.2.152
 - d. User Unit 3 (DVR) - IP address 192.168.2.153
 - e. User Unit 4 (CO-DVR) - IP address 192.168.2.154
3. Default IP addresses for various sub-units in AATV
 - a. Central Switch Unit - IP address 192.168.2.150
 - b. User Unit 1 (DOC) - IP address 192.168.2.151
 - c. User Unit 2 (DVR) - IP address 192.168.2.152
 - d. User Unit 3 (CO-DVR) - IP address 192.168.2.153

Notes: -

1. These IP addresses are suggested and used in the factory configured scenario.
2. Number of user units depends on the vehicle platform hence IP addresses will be in sequence from 192.168.2.151 onwards

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		Verified By	Prafulla Sarode	22.07.2026

8 Acceptance/ Performance Tests

8.1 Visual Inspection

S. No.	Description
1	The units / sub-units are labelled with proper name plate.
2	There is no physical damage, and painting is proper without any scratches.
3	OLED Display and Connectors presence to be verified.

Table 9: Visual inspection

8.2 Dimensional Inspection

S. No.	Parameter	Expected result
1	Dimensions of Central Switch Unit	As per Table 1
2	Dimensions of User Unit	As per Table 2
3	Dimensions of QWERTY Keyboard Unit	As per Table 3

Table 10: Dimensional Inspection

Note: Record the Result in Table for each Unit as shown in the table below.

A. Central Switch Unit Physical Inspection Record Table

S. No	Details of inspection	Expected Result	Observed Result
1	Label present on back side of cabinet with proper serial number	YES/NO	
2	Unit is without any damage	YES/NO	
3	Painting is proper and there are no scratches	YES/NO	
4	Length	210 mm (± 1 mm)	
5	Height	197 mm (± 1 mm)	
6	Depth	86 mm (± 1 mm)	
7	Weight	2.250 $\pm$ 0.200 Kg	
8	Colour	Olive-Green	
9	Radio Connector	4 nos (R/A-1 to R/A4)	
10	ETH & Power Connector	5 nos (J1 to J5)	
11	SPE & Power Connector	1 no (J6)	
12	Input Power Connector	1 no (24V-DC)	
13	Power On/Off Switch	1 no (ON/OFF)	
15	MGMT ETH Connector	1 no (MGMT)	
16	Utility-1	0 no (UTILITY-1)	
17	Utility-2	1 no (UTILITY-2)	
18	Antenna Connector	1 no (ANT)	

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		Verified By	Prafulla Sarode	22.07.2026

Table 11: iVIS Central Switch Unit - Physical Inspection Record

B. User Unit Physical Inspection Record Table

S. No	Details of inspection	Expected Result	Observed Result
1	Label present on back side of cabinet with proper serial number	YES/NO	
2	Unit is without any damage	YES/NO	
3	Painting is proper and there are no scratches	YES/NO	
4	Length	147 ± 1 mm	
5	Length with mounting cuts	180 ± 1 mm	
6	Height without protective hinges	142 ± 1 mm	
7	Height with protective hinges	157 ± 1 mm	
8	Depth	84 ± 1 mm	
9	Weight	1.500 ± 0.100 Kg	
10	Colour	Olive-Green	
11	Headset Connector	1 no (Headset)	
12	ETH & Power Connector	1 no (ETH)	
13	SPE & Power Connector	1 no (SPE)	
14	QWERTY Keyboard Connector	1 no (KBRD)	
15	Input Power Connector	0 no (PWR)	
16	Volume Knob	1 no	
17	Volume Knob Position	11 positions	
18	Mode Knob	1 no	
19	Mode Knob position	6 positions	
20	OLED Display	1 no	
21	Built-In Keypad Buttons	8 no's	

Table 12: iVIS User Unit - Physical Inspection Record

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

C. QWERTY Keyboard Physical Inspection Record Table

S. No	Details of inspection	Expected Result	Observed Result
1	Label present on back side of cabinet with proper serial number	YES/NO	
2	Unit is without any damage	YES/NO	
3	Painting is proper and there are no scratches	YES/NO	
4	Length	136.00 ± 0.50 mm	
5	Width	76.00 ± 0.50 mm	
6	Depth	20.00 ± 0.50 mm	
7	Color	Olive-Green	
8	Keyboard Buttons	49 Nos	
9	Weight	0.270 ± 0.050 kg	

Table 13: iVIS QWERTY Keyboard Unit - Physical Inspection Record

D. Stick Radio Dock Station Physical Inspection Record Table

S. No	Details of inspection	Expected Result	Observed Result
1	Label present on back side of cabinet with proper serial number	YES/NO	
2	Unit is without any damage	YES/NO	
3	Painting is proper and there are no scratches	YES/NO	
4	Length	150.00 ± 0.50 mm	
5	Height	90.00 ± 0.50 mm	
6	Depth	85.00 ± 0.50 mm	
7	Color	Olive-Green	
8	Input Power Connector	1 Nos	
9	Weight	0.270 ± 0.050 kg	

Table 14: Stick Radio Dock Station - Physical Inspection Record

 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
		Prepared By Shreyash K.	Shreyash K.	22.07.2026
		Verified By Prafulla Sarode	Prafulla Sarode	22.07.2026

8.3 Electrical Checks

8.3.1 Power ON and Power Consumption Check

A. Objective

To test the power ON status of the UUT and to record power consumption.

B. Test Procedure

1. Make the connection in CMTV configuration as Figure 18.
2. Connect the power supply to the Central Switch Unit.
3. Connect DMM for current measurement as shown in figure

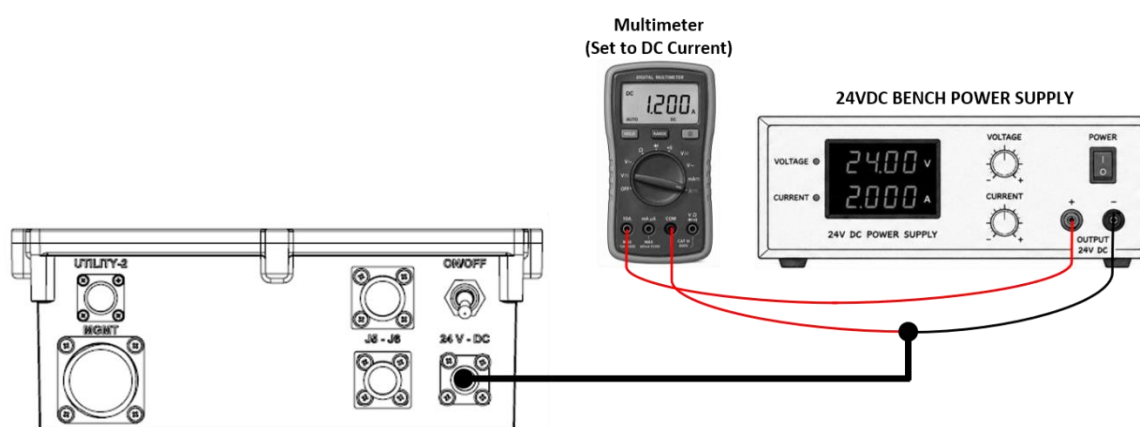


Figure 12: Multimeter Connection – CSU

4. Switch ON the 'ON/OFF' Switch of power source. **Check Result-1.**
5. Wait for the system to be initialized completely.
6. Measure the Voltage at the source and Current using Multi Meter. **Check Result-2**

C. Expected Result:

Result-1: The system shall power on and start in 30 seconds max.

Result-2: Voltage reading shall be DC 24V $\pm 5\%$, Current shall be less than 2 Amp for entire system.

D. Table of Record - Power ON and Power Consumption Check

Reference: ATP Para 8.3.1			
S. No	Required Result	Actual Result	Pass/Fail
Result-1	All units shall power ON within 30 seconds		
Result-2	Voltage shall be 24V DC $\pm 5\%$		
	Current shall be Less than 2 Amp.		
Observations (If any)			

Table 15: Record Table - Power ON Time and Power Consumption

 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
		Prepared By Shreyash K.	Shreyash K.	22.07.2026
		Verified By Prafulla Sarode	Prafulla Sarode	22.07.2026

8.3.2 Reverse Voltage Protection Test Procedure

A. Objective

To verify that the iVIS is protected against reverse polarity connection of the DC input supply and that no permanent damage occurs to the equipment.

B. Procedure

- 1 Make the system connection in CMTV configuration as per Figure 18.
- 2 Inspect the iVIS system for any visible physical damage before testing.
- 3 Connect the DCH to the DC power supply with correct polarity.
- 4 Positive (+) → Red terminal
- 5 Negative (−) → Black terminal
- 6 Switch ON the supply and verify
- 7 System powers ON normally
- 8 All functional indications are normal.
- 9 Switch OFF the power supply.
- 10 Reverse the DC input connections as shown in figure below.

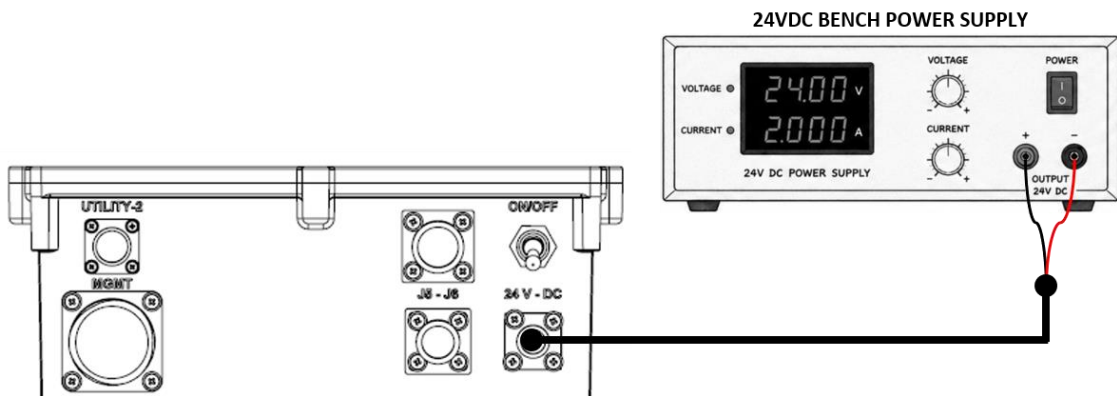


Figure 13: Reverse Voltage Connection – CSU

- 11 Positive (+) of supply → Black terminal of iVIS Cable
- 12 Negative (−) of supply → Red terminal of iVIS Cable
- 13 Switch ON the supply and verify for 30 seconds.

C. Expected Result

Observe the iVIS during reverse polarity application

- No smoke is generated. No abnormal heating occurs.
- No spark or burning smell is observed.
- Protective device (reverse polarity protection circuit) operates correctly.
- Equipment remains OFF or protected as designed
- Disconnect the reverse polarity supply.
- Reconnect the power supply with the correct polarity of wires.
- Switch ON the equipment.

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		Prepared By Shreyash K.	Shreyash K.	22.07.2026
		Verified By Prafulla Sarode	Prafulla Sarode	22.07.2026

- The DCH system is power on normally and working.

D. Table of Record – Reverse Voltage Protection

Reference: ATP Para 8.3.2			
S. No	Required Result	Actual Result	Pass/Fail
Result-1	No Unit should power on during reverse polarity input voltage		
Result-2	System should power on normally when correct polarity voltage is applied		
Observations (If any)			

Table 16: Record Table – Reverse Voltage Protection

8.3.3 Short Circuit Protection Test Procedure

A. Objective

To verify that the DCH System is capable of withstanding an accidental short circuit.

B. Procedure

- Make the system connection in CMTV configuration as per Figure 18.
- Power ON the System and confirm that all system functions are operating normally
- Connect the SPE & Power (Test cable) Cable on the SPE port of CSU.
- Intentionally short the +V (Red wires) and -V (Black wires) at the open-ended side as shown in figure below

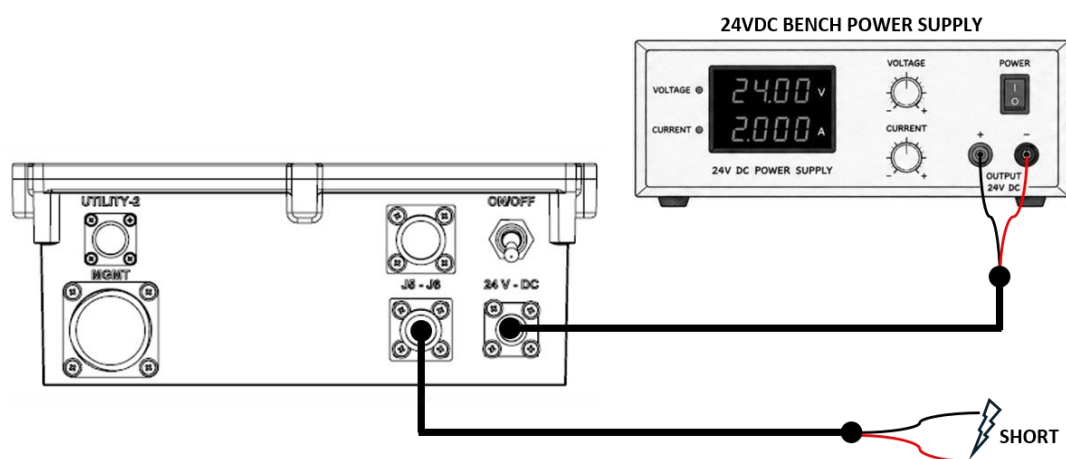


Figure 14: Short Circuit Connection - CSU

- Maintain the short circuit for 30 seconds
- Observe the User Unit display are off during the short-circuit condition.
- Remove the short circuit from the SPE Power Cable.

C. Expected Result

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	Verified By		Prafulla Sarode	22.07.2026

Observe the system during the short-circuit condition

- No smoke is generated.
- No burning smell is noticed.
- No abnormal heating occurs.
- Remove the short circuit from the SPE Power Cable.
- The protection circuit shall automatically restore normal operation
- The iVIS System shall resume normal functionality.
- No hardware replacement or repair shall be required.
- All communication and operational functions shall perform as specified.

D. Table of Record – Short Circuit Protection

Reference: ATP Para 8.3.3			
S. No	Required Result	Actual Result	Pass/Fail
Result-1	System shutdowns immediately at Short Circuit		
Result-2	System auto recovers as soon as short circuit is removed.		
Observations (If any)			

Table 17: Record Table – Short Circuit Protection

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		Prepared By Shreyash K.	Shreyash K.	22.07.2026
		Verified By Prafulla Sarode	Prafulla Sarode	22.07.2026

8.3.4 Power ON and Power Consumption Check of Stick Radio Dock Station

A. Objective

To test the power ON status of the UUT and to record power consumption.

B. Test Procedure

1. Connect the power supply to the Stick Radio Dock Station.
2. Connect DMM for current measurement as shown in figure below.

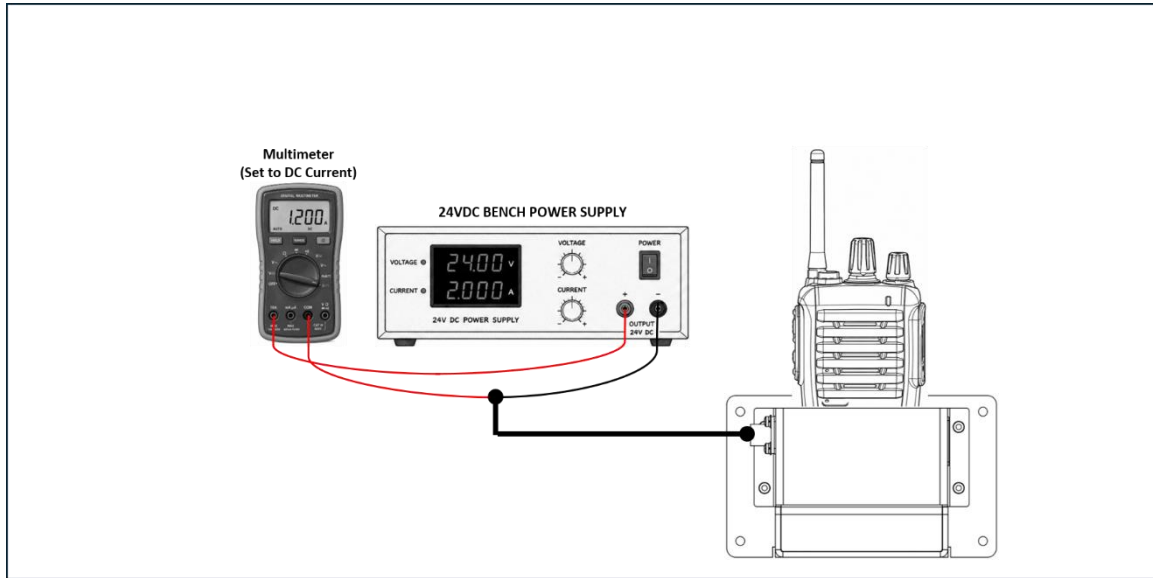


Figure 15: Multimeter Connection – Stick Radio Dock Station

3. Turn on the power supply.
4. Measure the Voltage at the source and Current using Multimeter.

C. Expected Result:

Result-1: Voltage reading shall be DC 24V $\pm 5\%$, Current shall be less than 0.165 Amp for entire system.

D. Table of Record - Power Consumption Check of Stick Radio Dock Station

Reference: ATP Para 8.3.4			
S. No	Required Result	Actual Result	Pass/Fail
Result-1	Voltage shall be 24V DC $\pm 5\%$		
	Current shall be Less than 0.165 Amp ± 0.02 Amp		
Observations (If any)			

Table 18: Record Table - Power ON Time and Power Consumption

 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
		Prepared By	Shreyash K.	22.07.2026
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8.3.5 Reverse Voltage Operation Test Procedure

A. Objective

To verify that the Stick Radio Dock Station is protected against reverse polarity connection of the DC input supply and that no permanent damage occurs to the equipment.

B. Procedure

- 1 Connect the Stick Radio Dock Station to the DC power supply with correct polarity.
- 2 Positive (+) → Red terminal
- 3 Negative (–) → Black terminal
- 4 Switch ON the supply and verify
- 5 Stick Radio Dock Station powers ON normally
- 6 Switch OFF the power supply.
- 7 Reverse the DC input connections as shown in figure below.

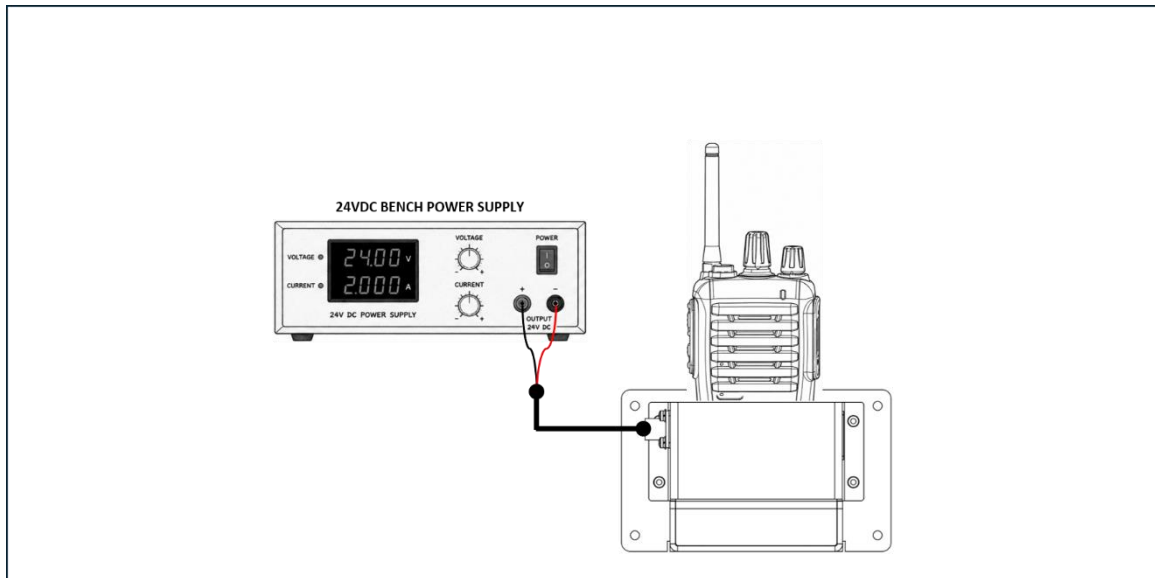


Figure 16: Test Setup - Reverse Voltage Operation – Stick Radio Base Station

- 8 Positive (+) of supply → Black terminal of Power Cable
- 9 Negative (–) of supply → Red terminal of Power Cable
- 10 Switch ON the supply and verify.
- 11 Stick Radio Dock Station powers ON normally

C. Expected Result

Result-1: In both the polarity(Correct and Reversed) Stick Radio Dock Station Should Operate Normally. No Function should be affected in both the polarity

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
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	Verified By		Prafulla Sarode	22.07.2026

D. Table of Record – Reverse Voltage Operation

Reference: ATP Para 8.3.5			
S. No	Required Result	Actual Result	Pass/Fail
Result-1	No Unit should power on during reverse polarity input voltage		
Result-2	System should power on normally when correct polarity voltage is applied		
Observations (If any)			

Table 19: Record Table – Reverse Voltage Operation – Stick Radio Base Station

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		Verified By 	Prafulla Sarode	22.07.2026

8.4 Test Setup for AATV

To conduct functional test, version verification test, interfaces verification tests, a test setup for AATV as depicted in Figure 17.

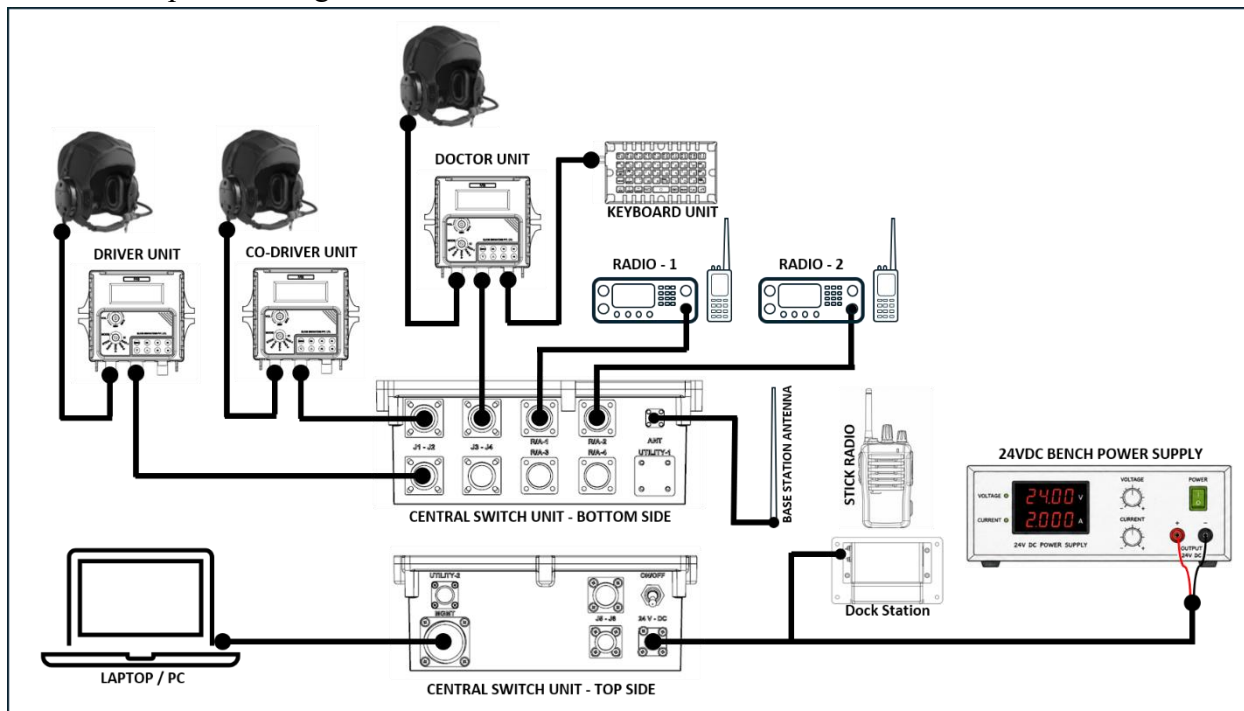


Figure 17: iVIS Test Setup – AATV

8.5 Test Setup for CMTV

To conduct functional tests, version verification test, interfaces verification tests, a test setup for CMTV as depicted in Figure 18.

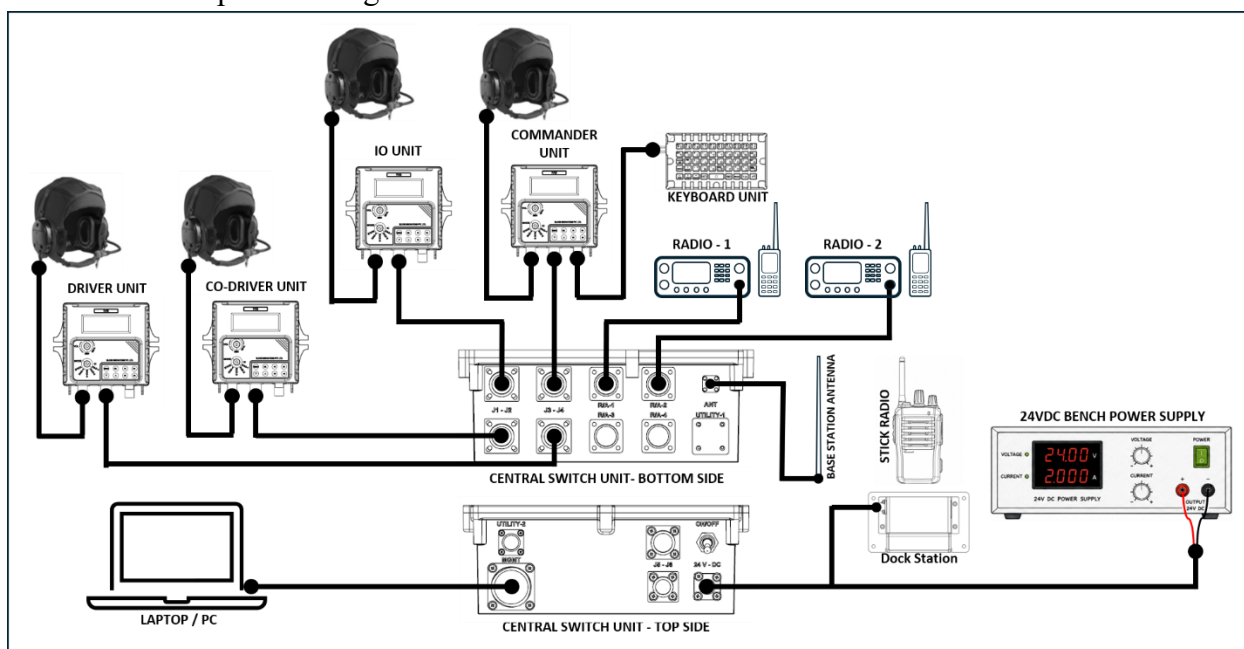


Figure 18: iVIS Test Setup – CMTV

 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

8.6 System Power on Time and Connection Test

A. Objective

To verify that the Central Switch Unit and all associated User Units power on within 30 seconds after being supplied with 24V DC and establish a functional Ethernet (ETH) connection both with each other and with the connected PC. This test ensures that all units initialize properly, communicate over the network, and respond to ping requests, confirming system readiness and robust inter-unit connectivity.

B. Procedure

- 1 Make the system connection in CMTV configuration as per Figure 18.
- 2 Make sure On/Off switch on iVIS CSU is at Off Position. and turn on the 24VDC Bench Power Supply.
- 3 Open clock on the Laptop/PC. And goto stopwatch section.
- 4 Turn ON the On/Off Switch on the Central Switch Unit and simultaneously turn start the stopwatch on the laptop.
- 5 Wait for the Home screen to appear on the User Unit
- 6 As soon as all error/warning Notifications disappear stop the stopwatch. And Note the time.

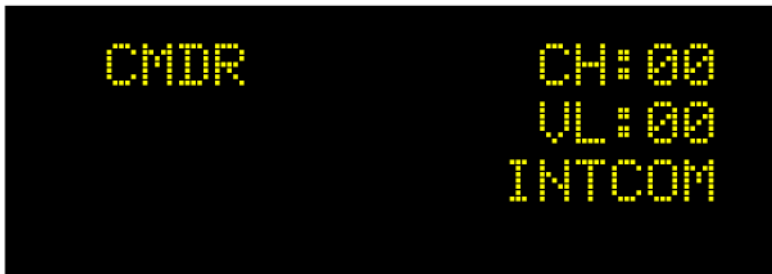


Figure 19: iVIS UU Display - Home Screen with no warnings



Figure 20: iVIS UU Display - Home Screen with DVR down warning

C. Expected Result

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		Verified By	Prafulla Sarode	22.07.2026

Result-1: Central Switch Unit and all User Units should power ON within **30 seconds** and display their respective configured titles on the home screen/page as shown in figures below.



Figure 21: iVIS UU Display - Home Screen – CMDR

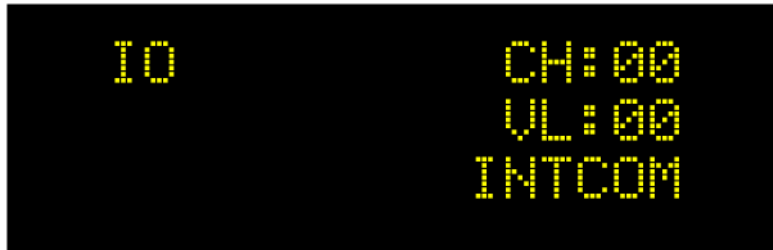


Figure 22: iVIS UU Display - Home Screen – IO



Figure 23: iVIS UU Display - Home Screen – DVR



Figure 24: iVIS UU Display - Home Screen – CO-DVR

Note: Record the Result in Table below

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

Reference: ATP Para 8.6			
S. No.	Required Result	Actual Result	Pass/Fail
Result-1 for UUT	User Unit - 1 (CMDR) should be powered on within 30 seconds. And should display home screen as shown in Figure 21		
	User Unit - 2 (IO) should be powered on within 30 seconds. And should display home screen as shown in Figure 22		
	User Unit - 3 (DVR) should be powered on within 30 seconds. And should display home screen as shown in Figure 23		
	User Unit - 4 (CO-DVR) should be powered on within 30 seconds. And should display home screen as shown in Figure 24		
	Any additional user unit/units should be powered on within 30 seconds.		
Observations (If any)			

Table 20: System Power on Time and Connection Test

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

8.7 Interface verification for User Unit

For User Unit interface verification,

1. Make the system connection in CMTV configuration as per Figure 18.
2. Power ON the Central Switch Unit using 24V DC.
3. Wait for 30 seconds for the system to be initialized completely.
4. After initialization of the user unit, the display of each unit will display its configured name such as “CMDR”, “IO”, “DVR” and “CO-DVR”.

8.7.1 Knob Test

Steps:

1. On the Home Screen, rotate the Volume knob from 0th position to 10th position.
2. Rotate the Volume knob back from 10th position to 0th position.
3. Rotate the Operation Mode knob from 1st position to 6th position.
4. Rotate the Operation Mode knob back from 6th position to 1st position.

Expected Results:

1. The volume level “VL:” is displayed and updates correctly as the knob rotates. At 0th position VL:00 and at 10th position VL:10. And all the values in between should be displayed respectively in sequence.



Figure 25: iVIS UU Display - Home Screen - CMDR - VL:00



Figure 26: iVIS UU Display - Home Screen - CMDR - VL:10

2. The Mode “CH:” is displayed and updates correctly as the knob rotates. At 0th position CH:00 and at 6th position CH:05. And all the values in between should be displayed respectively in sequence.

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		Verified By	Prafulla Sarode	22.07.2026



Figure 27: iVIS UU Display - Home Screen - CMDR – CH:00

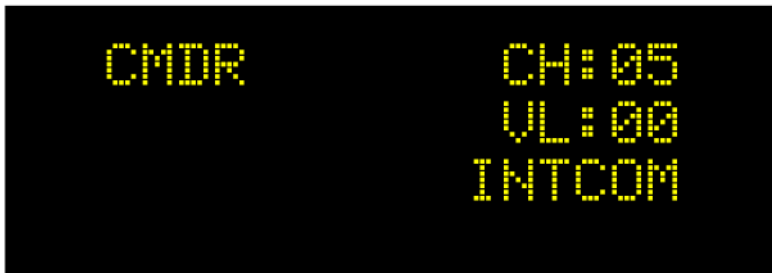


Figure 28: iVIS UU Display - Home Screen - CMDR - CH:06

- Rotating back should display the reverse sequence.

8.7.2 Display Test

Steps:

- Click the "Menu" button.
- Scroll down to "Diagnostics" and click "Diagnostics".
- Click "BITE".
- Click "Display Test".
- Click "OK" repeatedly to cycle through display test stages.

Expected Results:

- 1st OK: All display matrix characters turn ON (screen fully lit).
- 2nd OK: All display matrix characters turn OFF.
- 3rd OK: All display matrix characters change to '8'.
- 4th OK: Display test finishes and returns to normal screen.



Figure 29: iVIS UU Display - BYTE - Display Test Screen Fully On

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		Verified By	Prafulla Sarode	22.07.2026

iVIS (UCZN10331)

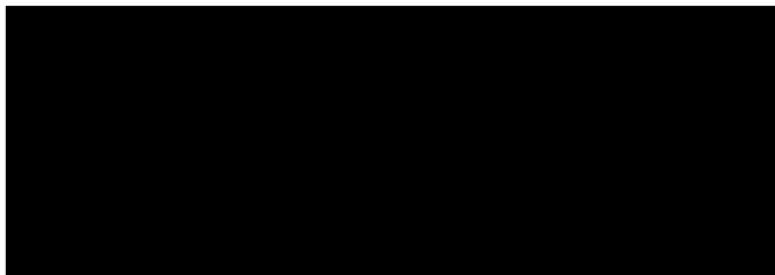


Figure 30: iVIS UU Display - BYTE - Display Test Screen Fully Off

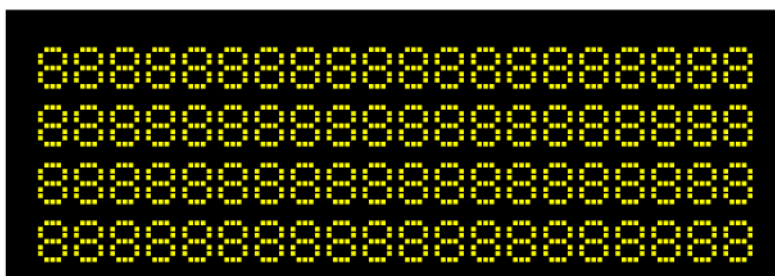


Figure 31: iVIS UU Display - BYTE - Display Test All Characters '8'

8.7.3 Built-in Keypad Test

Steps:

1. Click the "Menu" button.
2. Scroll down to "Diagnostics" and click "Diagnostics".
3. Click "BITE".
4. Click "Keypad Test".
5. Press all keypad buttons one by one in random order. Which are on User Unit (F1,F2,F3,F4 up/down key and ok). Press Menu key to Exit.

Expected Results:

1. Each pressed button label is displayed correctly on the Unit screen. As shown in the figure below for F3 key press



Figure 32: iVIS UU Display - BYTE – Keypad Test

2. After pressing all buttons, the test finishes automatically.

8.7.4 PTT Test

Steps:

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		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

1. Click the "Menu" button.
2. Scroll down to "Diagnostics" and click "Diagnostics".
3. Click "BITE".
4. Click "PTT Test".
5. Display will show PTT number and release
6. Press PTT in momentary position(forward position) and release it.
7. Observe the result 1
8. Press PTT in latching position(backward position).
9. Observe the result 2
10. Released PTT from latching position(backward position) by moving it to center position.
11. Observe the result 3
12. Press Menu key to back step.

Expected Results:

1. "PTT 1:" will change from RELEASED to PRESSED when forward PTT is pressed and it will change from PRESSED to RELEASED when forward PTT is released.



Figure 33: iVIS UU Display - BYTE - PTT Test PTT1 Pressed

2. "PTT 2:" will change from RELEASED to PRESSED when backward PTT is pressed.
3. "PPT2:" will change from PRESSED to RELEASED when backward PTT is released to center position.

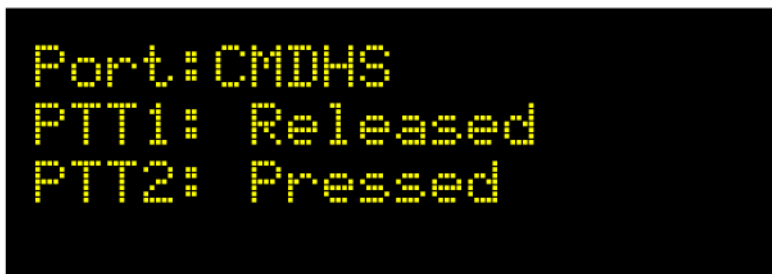


Figure 34: iVIS UU Display - BYTE - PTT Test PTT2 Pressed

8.7.5 Headset Speaker Test

Steps:

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		Verified By	Prafulla Sarode	22.07.2026

1. Connect the headset to the User Unit.
2. Set the position of volume knob to MIN position by turning the knob anticlockwise
3. Turn volume knob clockwise
4. Listen to tone and increasing volume in clockwise directions
5. Turn volume knob anticlockwise.
6. Listen to tone and decreasing volume in anticlockwise directions
7. Turn the Mode Knob in any direction
8. Listen to Audio Channel Prompts

Expected Results:

1. Tone with increasing volume shall be heard when volume knob is turned in clockwise direction.
2. Tone with decreasing volume shall be heard when volume knob is turned in anticlockwise direction.
3. Voice Prompt will be announced according to current channel.

8.7.6 Headset Mic Test**Steps:**

1. Click the "Menu" button.
2. Scroll down to "Diagnostics" and click "Diagnostics".
3. Click "BITE".
4. Click "MIC Test".
5. Click "Start Test".
6. Press PTT, and record ok to stop recording is shown in the display. Now, speak into the mic to record the audio message, after recording release the PTT.
7. Press OK to "Play Recording" is displayed
8. Press OK to listen to playback
9. Press Ok to continue.
10. MIC Test Complete shown in display.
11. Press MENU/OK to finish test

Expected Results:

1. Audio is recorded when PTT is pressed.
2. Playback matches recorded speech through the headset speaker.
3. Test finishes upon clicking "Finish Test".

8.8 Keyboard test

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		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

8.8.1 Objective

To check QWERTY keyboard functions.

8.8.2 Procedure

- 1 Make the system connection as per Figure 18.
- 2 Connect the keyboard to any of the User units
- 3 Click the "CMD" button.
- 4 Scroll down to "Diagnostics" and click "Diagnostics".
- 5 Click "BITE".
6. Click "Keyboard Test".
7. Press all keyboard buttons one by one in random order. i.e. follow instructions displayed on display screen

8.8.3 Expected Result

Result-1: Each pressed button label is displayed correctly on the User Unit screen. Except FN (Function) and MM (Multimedia) key

8.9 Interface verification for Central Switch Unit**8.9.1 Radio Interface verification**

1. Connect the system components in CMTV configuration as shown in Figure 18.
2. Connect the radio Set on Radio port one by one
3. Power ON the system.
4. Allow 30 seconds for the system to start fully.
5. Set the Reference Radio Handheld (e.g., Motorola Radio Handheld) to the same channel as Radio connected to the iVIS system.
6. On any one of the User Units, select RAD-1 (respective radio)
7. Reference Radio Transmission Test
 - Press and hold the PTT button on the Reference Radio.
 - Speak into the Reference Radio microphone.
 - Observe Result-1.
8. User Unit Transmission Test
 - Press and hold the PTT button on the headset connected to the User Unit.
 - Speak into the headset microphone.
 - Observe Result-2.
9. Repeat for All Radios

Repeat the procedure for Radio-2, Radio-3 and Radio-4,

Expected Results for Radio Interface verification

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Result-1: Audio from the Reference Radio should be clearly audible through the headset speaker connected to the User Unit.

Result-2: Audio from the User Unit headset microphone should be clearly audible on the Reference Radio.

8.9.2 Headset communication Verification

This procedure verifies communication between User Units connected to the Central Switch Unit (CSU). Each User Unit is connected via connecting cable and equipped with a headset, enabling internal communication (IC mode).

1. Connect all User Units to the CSU using connecting cables in CMTV configuration as shown in Figure 18.
2. Power ON the system.
3. Allow 30 seconds for the CSU and all connected User Units to initialize completely.
4. Ensure that a functional headset is securely connected to each of the User Units.
5. On each User Unit, select IC mode (intercom mode) to enable internal communication between users.

8.9.3 Voice Communication Test (IC Mode)

- From User Unit 1:
 - Press and hold the PTT button on the headset.
 - Speak into the headset microphone.
 - Confirm that audio is clearly received on headset speakers of other User Units.
- From User Unit 2:
 - Press and hold the PTT button on the headset.
 - Speak into the headset microphone.
 - Confirm that audio is clearly received on headset speakers of other User Units
- Repeat the test for other remaining User Units as applicable, confirming successful communication with all other units.

Expected Results for Headset communication Verification

- **Result-1:** Audio transmitted from one User Unit's headset microphone in IC mode should be clearly audible through the headset speakers of all other connected User Units.
- **Result-2:** There should be no audio distortion, delay, or loss in inter-unit communication.

8.9.4 Stick Radio Interface verification

1. Connect the system components as shown in Figure 17 or Figure 18.

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

2. Power ON the Central Switch Unit using a 24V DC power source.
3. Allow 30 seconds for the system to initialize fully.
4. On any one of the User Units, select Stick.
5. Stick Radio Transmission Test
 - Press and hold the PTT button on the Stick Radio.
 - Speak into the Reference Radio microphone.
 - Observe Result-1.
6. User Unit Transmission Test
 - Press and hold the PTT button on the headset connected to the User Unit.
 - Speak into the headset microphone.
 - Observe Result-2.

Expected Results for Radio Interface verification

- **Result-1:** Audio from the Stick Radio should be clearly audible through the headset speaker connected to the User Unit.
- **Result-2:** Audio from the User Unit headset microphone should be clearly audible on Stick Radio.

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		Verified By Prafulla Sarode	Prafulla Sarode	22.07.2026

8.9.5 Radio Messaging Test

A. CMTV Setup and Settings for Radio Messaging.

1. Make system connection in CMTV configuration as shown in figure below.

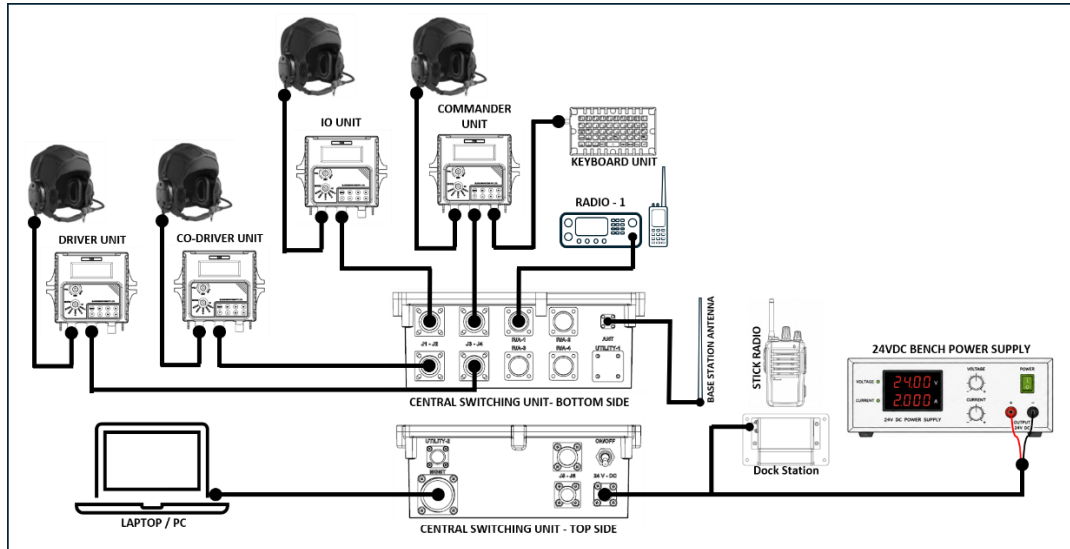


Figure 35: iVIS Test Setup – CMTV (Radio Messaging)

2. On any User Unit in CMTV Setup press MENU.
3. Scroll down and click on Settings.
4. Enter Password “F1 F2 F3 F4”
5. Scroll down and Click on Message Method
6. Make sure Radio 1’s Message Source is Intercom and Message Mode is Voice as shown in Figure 36.



Figure 36: iVIS UU Display – Message Method

7. If Message Source or Message Mode is incorrect, then it can be changed using OK and Up/Down arrow keys
8. In settings, locate Vehicle ID and press OK.
9. Enter Vehicle ID as “CMTV-1”
10. In settings, locate message destination and press OK
11. Select Add Destination, and enter “DEMO-11”
12. In settings, locate select destination and press OK
13. In Radio 1 click OK and select “DEMO-11”.

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		Verified By Prafulla Sarode	22.07.2026	

B. TEST Setup and Settings for Radio Messaging.

1. Make system connection in CMTV configuration as shown in figure below.

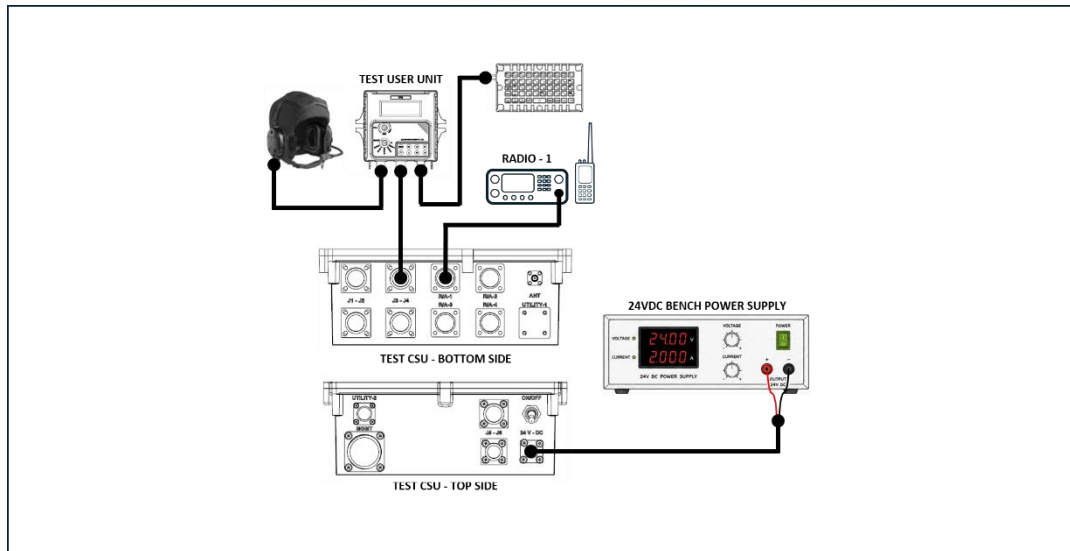


Figure 37: iVIS Test Setup – DEMO (Radio Messaging)

2. On Demo User Unit in DEMO Setup press MENU.
3. Scroll down and click on Settings.
4. Enter Password “F1 F2 F3 F4”
5. Scroll down and Click on Message Method
6. Make sure Radio 1’s Message Source is Intercom and Message Mode is Voice as shown in Figure 36.
7. If Message Source or Message Mode is incorrect, then it can be changed using OK and Up/Down arrow keys
8. In settings, locate Vehicle ID and press OK.
9. Enter Vehicle ID as “DEMO-11”
10. In settings, locate message destination and press OK
11. Select Add Destination, and enter “CMTV-1”
12. In settings, locate select destination and press OK
13. In Radio 1 click OK and select “CMTV-1”.

C. Sending message from CMTV setup to DEMO setup

1. On any User Unit in CMTV Setup press MENU
2. Press Menu, scroll down to Functionality and click OK
3. Select Radio Message and Press on the Compose.
4. Type the message and click OK.
5. Select Radio 1 and click OK
6. Sent messages can be checked inside Outbox in Radio Messages.

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		Verified By	Prafulla Sarode	22.07.2026

D. Sending message from DEMO setup to TEST setup

1. On Demo User Unit in DEMO Setup press MENU.
2. Scroll down to Functionality and click OK
3. Select Radio Message and Press on the Compose.
4. Type the message and click OK.
7. Select Radio 1 and click OK
8. Sent messages can be checked inside Outbox in Radio Messages.

Expected Results for Radio Interface verification

- **Result-1:** Message received in DEMO setup from CMTV setup can be checked inside Inbox in Radio Message on the Demo User Unit of DEMO setup.
- **Result-2:** Message received in CMTV setup from DEMO setup can be checked inside Inbox in Radio Message on any User Unit of CMTV setup.

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	Verified By	Prafulla Sarode	22.07.2026	

9 Login procedure for verification of units

Open “PuTTY” software on Laptop and select the Telnet option, telnet port as 8889 and enter destination units IP Address as IP address of respective unit. As shown in **Error! Reference source not found.**

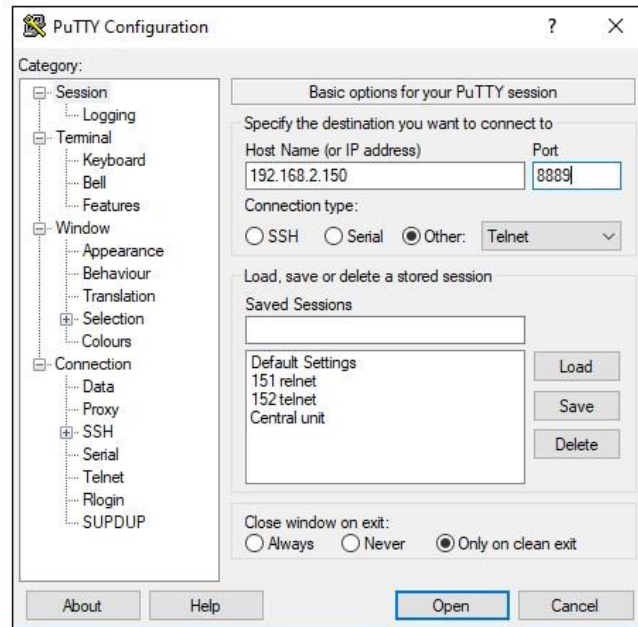
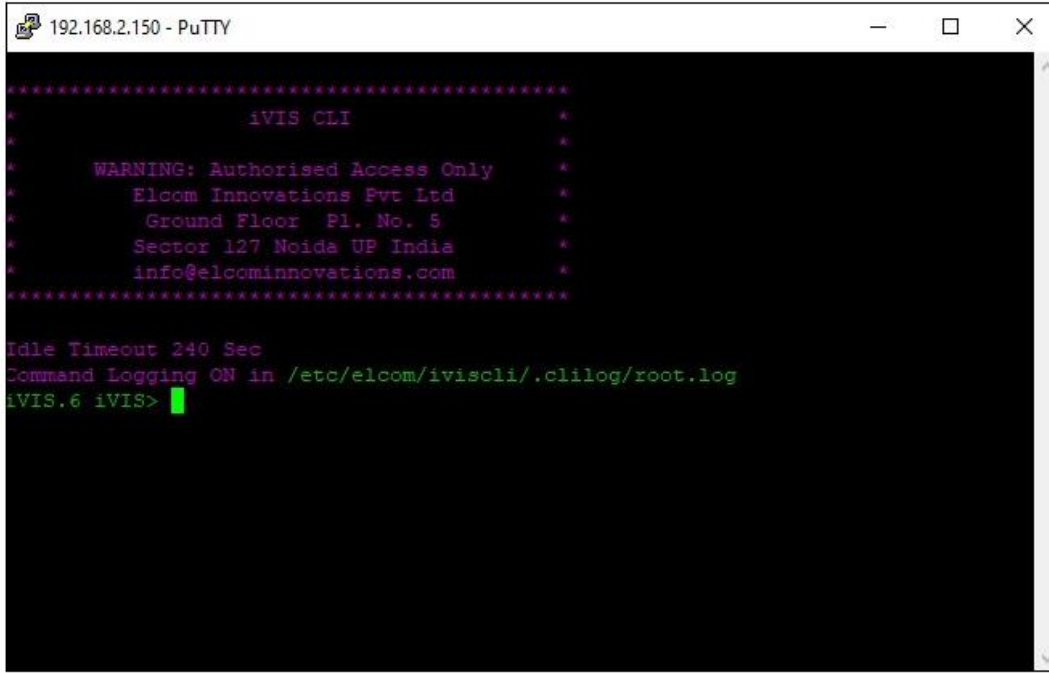


Figure 38: Login Procedure of Units

1. Press “Open”
2. System welcome prompt will be displayed as shown in **Error! Reference source not found.**

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		Verified By	Prafulla Sarode	22.07.2026



```

192.168.2.150 - PuTTY

*****
*                               *
*           iVIS CLI           *
*                               *
*  WARNING: Authorised Access Only  *
*    Elcom Innovations Pvt Ltd    *
*    Ground Floor Pl. No. 5      *
*    Sector 127 Noida UP India   *
*    info@elcominnovations.com   *
*                               *
* *****                      *
*                               *
Idle Timeout 240 Sec
Command Logging ON in /etc/elcom/iviscli/.clilog/root.log
iVIS.6 iVIS>
  
```

Figure 39: Welcome prompt of UnitVerification of versions

A. Objective

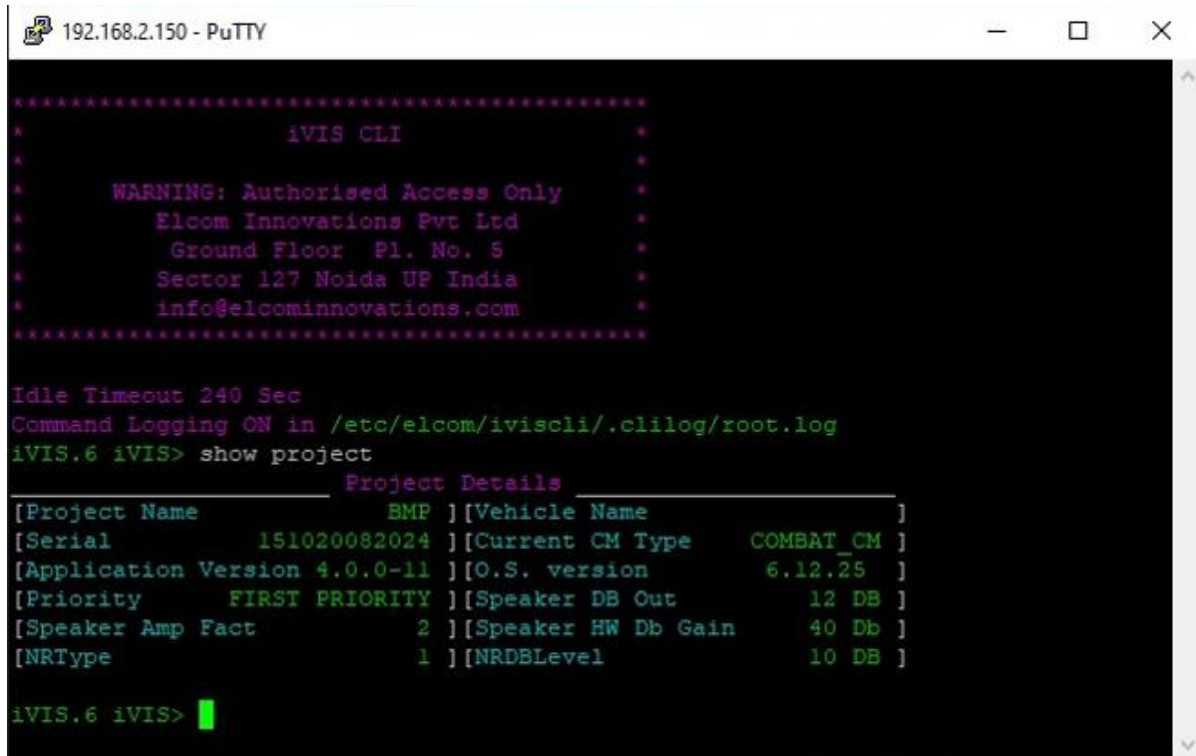
To check OS, application version and configuration present in each unit.

B. Procedure

- 1 Make the system connection as per Figure 17 or Figure 18.
- 2 Power ON the Central Switch Unit from 24V DC.
- 3 Check ETH connection from the PC:
- 4 PC set up/configuration:
 - (i) Configure the PC with IP address 192.168.2.101
 - (ii) Wait for 30 seconds for the system to initialize completely and observe the ping response for Central Switch Unit whose IP address is 192.168.2.150.
 - (iii) Login to Central Switch Unit and each user unit of which version is to be verified. For login refer to Login procedure as per para **Error! Reference source not found.** “Login procedure for verification of units”.
- 5 Issue command “**show project**” as shown in Figure 40.
- 6 Record the version details shown – Check Result-1.
- 7 Repeat the procedure for all other units.

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iVIS (UCZN10331)	Prepared By		Shreyash K.	22.07.2026
	Verified By		Prafulla Sarode	22.07.2026



```
*****
*                               *
*          iVIS CLI             *
*                               *
*  WARNING: Authorised Access Only *
*    Elcom Innovations Pvt Ltd   *
*    Ground Floor Pl. No. 5     *
*    Sector 127 Noida UP India  *
*    info@elcominnovations.com  *
*                               *
*****

Idle Timeout 240 Sec
Command Logging ON in /etc/elcom/iviscli/.clilog/root.log
iVIS.6 iVIS> show project

      Project Details
-----
[Project Name      BMP ] [Vehicle Name      ]
[Serial      151020082024 ] [Current CM Type    COMBAT_CM ]
[Application Version 4.0.0-11] [O.S. version      6.12.25 ]
[Priority      FIRST PRIORITY] [Speaker DB Out     12 DB ]
[Speaker Amp Fact      2 ] [Speaker HW Db Gain 40 Db ]
[NRType      1 ] [NRDBLevel         10 DB ]

iVIS.6 iVIS> █
```

Figure 40: Command for Versions check

C. Expected Result

Result-1 : OS & Application version shall be shown as per Table 21.

Note: Record the Result in Table below

Reference: ATP Para 9			
S. No.	Required Result	Result Observed	Pass/Fail
Result-1 for UUT	OS version	6.12.25	
	Application version	4.0.0-11 or higher	
Observations (If any)			

Table 21: System Versions

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		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

10 Voice Channel Verification

A. Objective

To check the Voice Channels speed

B. Procedure

- 1 Make the system connection as per Figure 17 or Figure 18.
- 2 Connect the PC/Laptop Ethernet port with the Central Switch Unit (MGMT Port) using a standard LAN cable.
- 3 Verify the network connectivity by pinging the IP address of the unit that needs to be tested.
- 4 Ensure that the ping response is successful.
- 5 Open the Wireshark application.
- 6 Select the Ethernet interface on Wireshark and double click on to start capturing packets as shown in Figure 41.

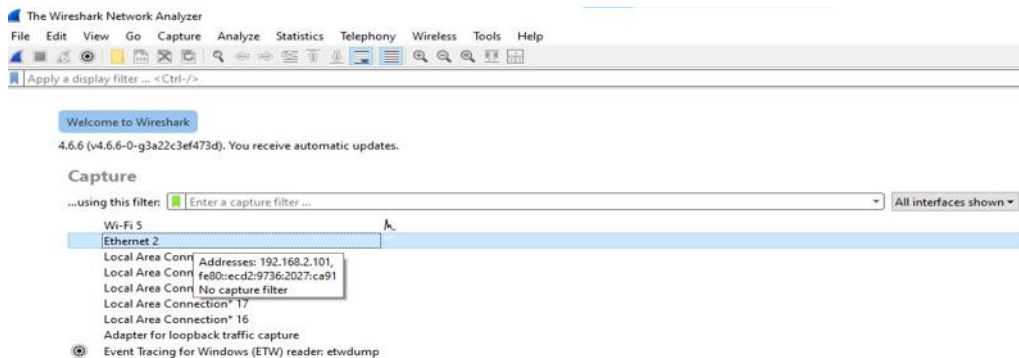


Figure 41: Wireshark Window for Ethernet selection

- 7 Capture window will open as shown in Figure 42.

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		Verified By Prafulla Sarode		

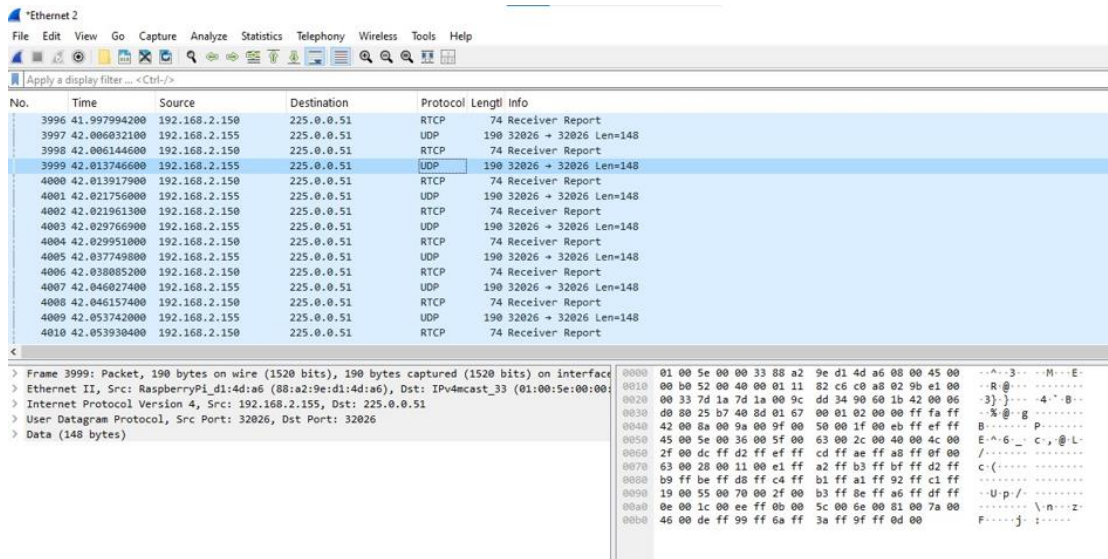


Figure 42: Wireshark Capture window

- 8 Press the PTT (Push-to-Talk) button of the Headset and speak into the headset for capturing the packets.
- 9 Release the PTT while capturing the data and stop the trace on Wireshark.
- 10 Select any UDP packet from the captured traffic in Wireshark Trace and click on it and select decode as show in Figure 43.

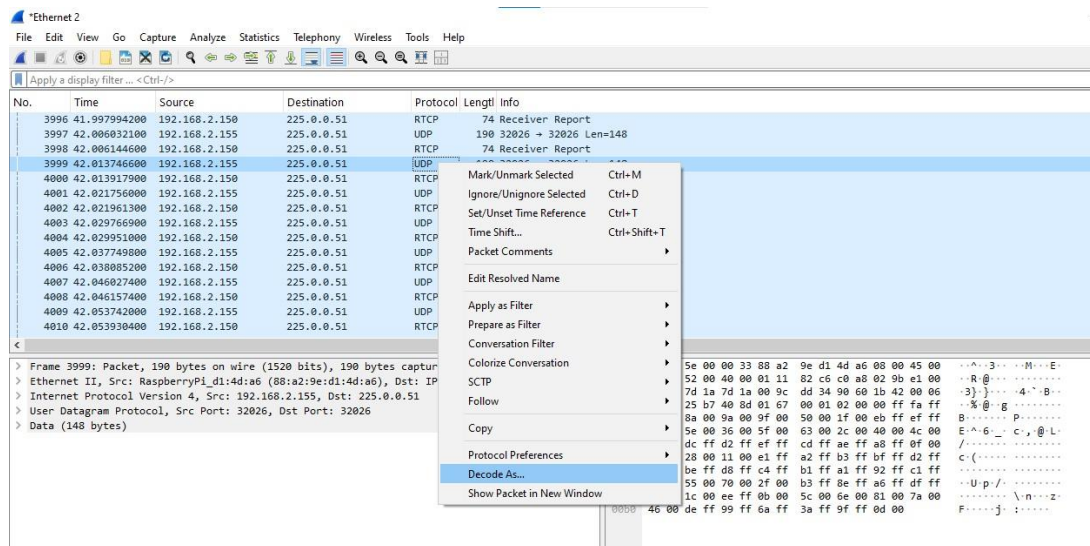


Figure 43: UDP Packet Decode as RTP

- 11 A pop-up window is open as shown below:

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		Verified By Prafulla Sarode	22.07.2026	

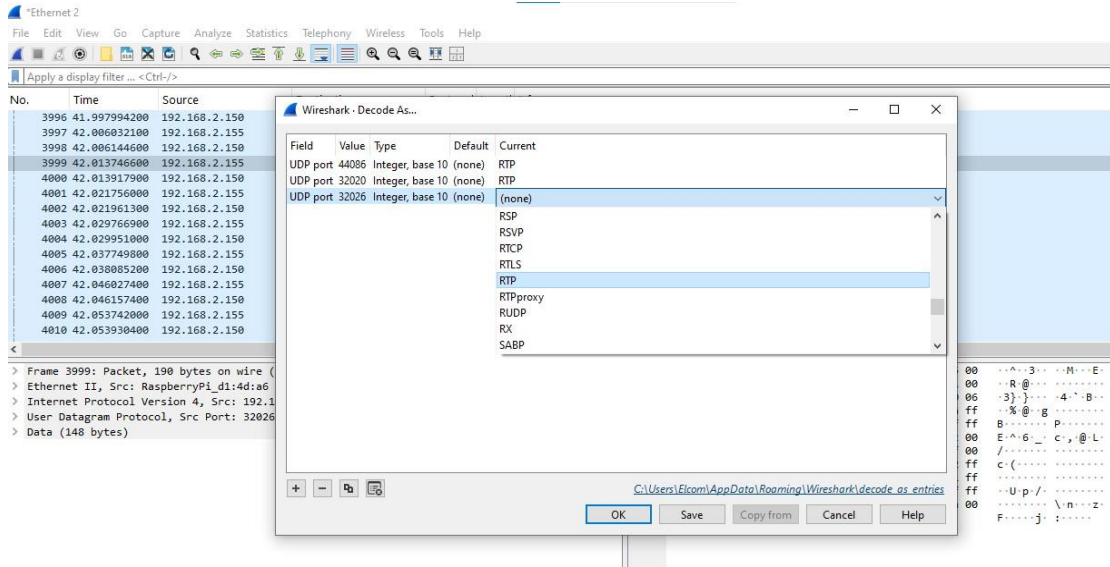


Figure 44: UDP Packet open as RTP

- 12 Click on the Data shown as UDP port 32026 (none) in above Figure 44 and converted into RTP.
- 13 Verify that the selected data packet is now decoded as RTP on Wireshark window as shown in Figure 45.

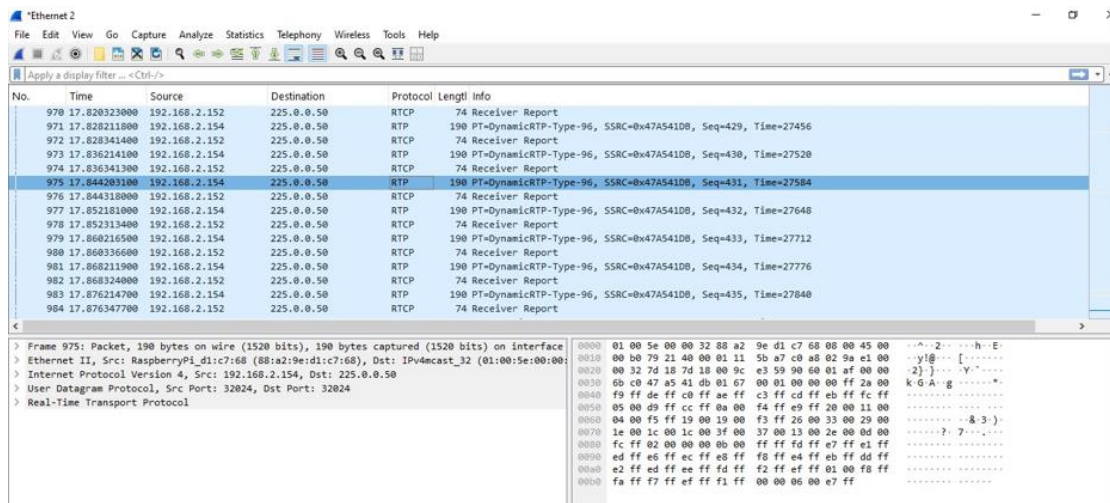


Figure 45: RTP Packet on Wireshark

- 14 Click on RTP packet and go to the Telephony menu at the top of the Wireshark window and click on RTP Player as shown in Figure 46.

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		Verified By	Prafulla Sarode	22.07.2026

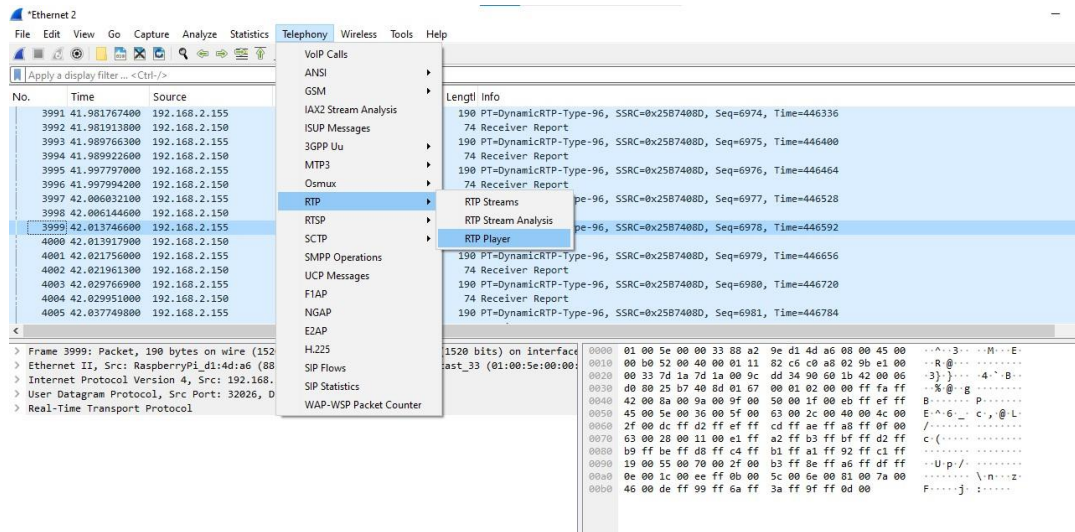


Figure 46: RTP Packet to RTP Player

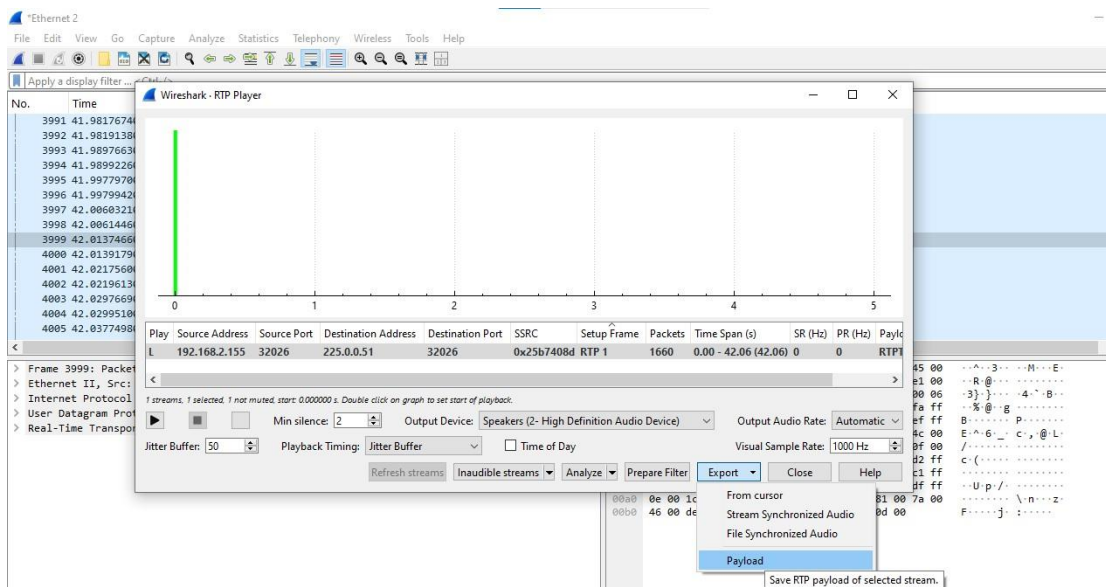


Figure 47: RTP Player to Payload

- 15 In the RTP Player window, select the required RTP data.
- 16 Click on Export tab and select Payload to save the data as shown in Figure 47.

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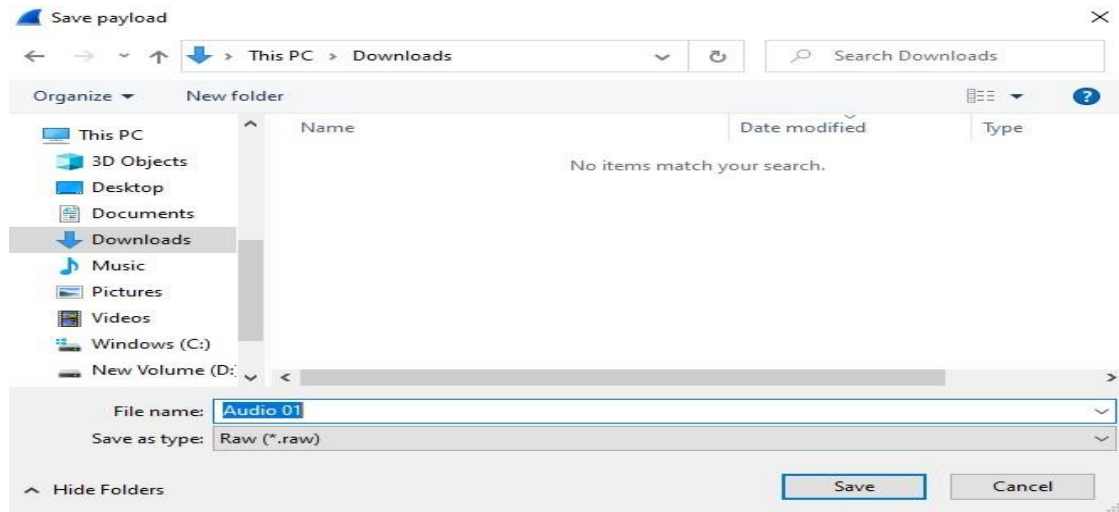


Figure 48: Save Data process

- 17 To save the data on Desktop select the file type as RAW, enter the desired file name, choose the save location, and save the file as shown in Figure 48.

Open the Audacity application.

- 1 Go to File Tab > Import > Raw Data, select the saved RAW file from file location, and click Open as shown in Figure 49.

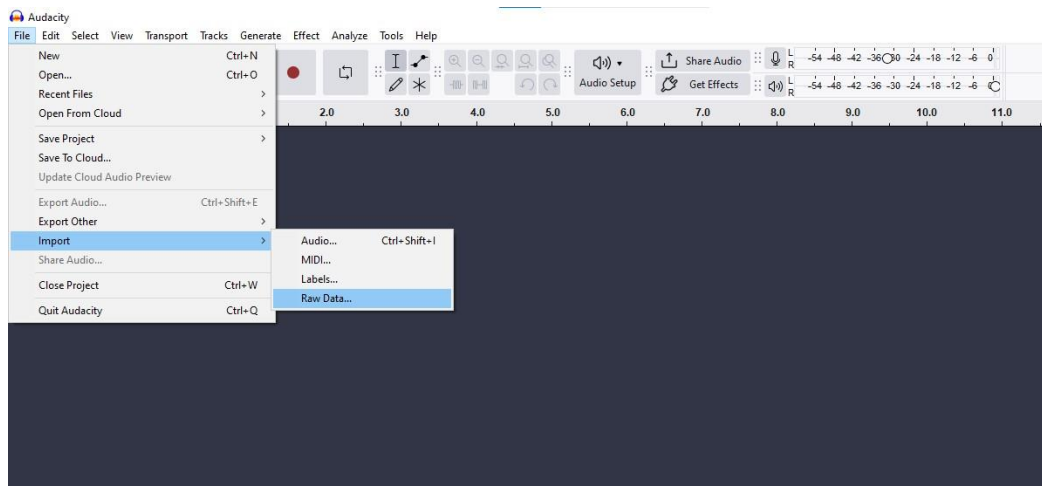


Figure 49: Import the saved RAW file

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		Verified By	Prafulla Sarode	22.07.2026

- In the import process another popup will display on the Audacity window as Import Raw Data: Audio 01 set Encoding to Signed 16-bit PCM and Sample Rate to 8000 Hz, then click Import as shown in Figure 50.

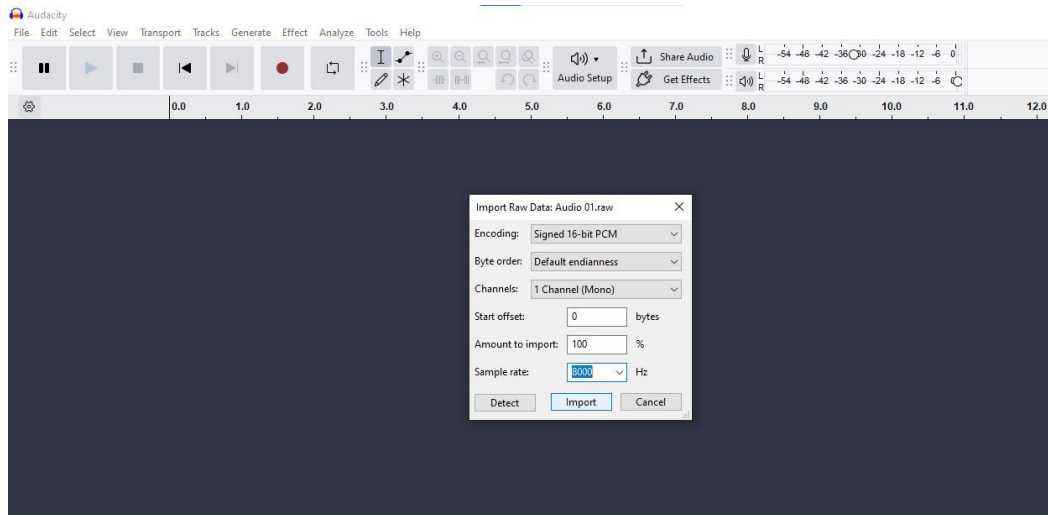


Figure 50: Sample Rate Selection

- The audio is imported, click the Play button to listen to the recorded voice as shown in Figure 51.

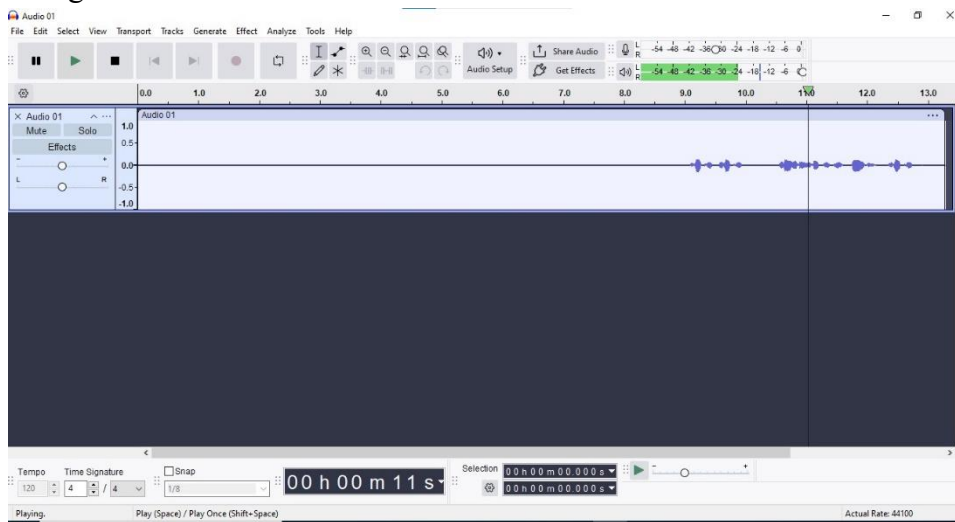


Figure 51: Play the Recoded Voice

C. Expected Result

The RAW audio file was successfully imported and verified in Audacity using Signed 16-bit PCM encoding at a sampling rate of 8000 Hz, resulting in an effective 128 kbps PCM voice stream. This exceeds the minimum specified requirement of 64 kbps PCM, thereby confirming compliance with the specification.

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11 GPS Interface Verification

A. Objective

To check the interface of GPS

B. Procedure

- 1 Connect the PC/Laptop serial port to the Central Switch Unit using a Serial Cable (connect to Utility-2 Port).
- 2 Open PuTTY application and select the serial connection type.
- 3 The following setting is entered as mentioned below in the PuTTY application.
 - (a) Serial Line is the COM port of PC/Laptop (e.g., COM1, COM2, COM3) as shown in Figure 52.
 - (b) Speed (Baud Rate) selected 57600

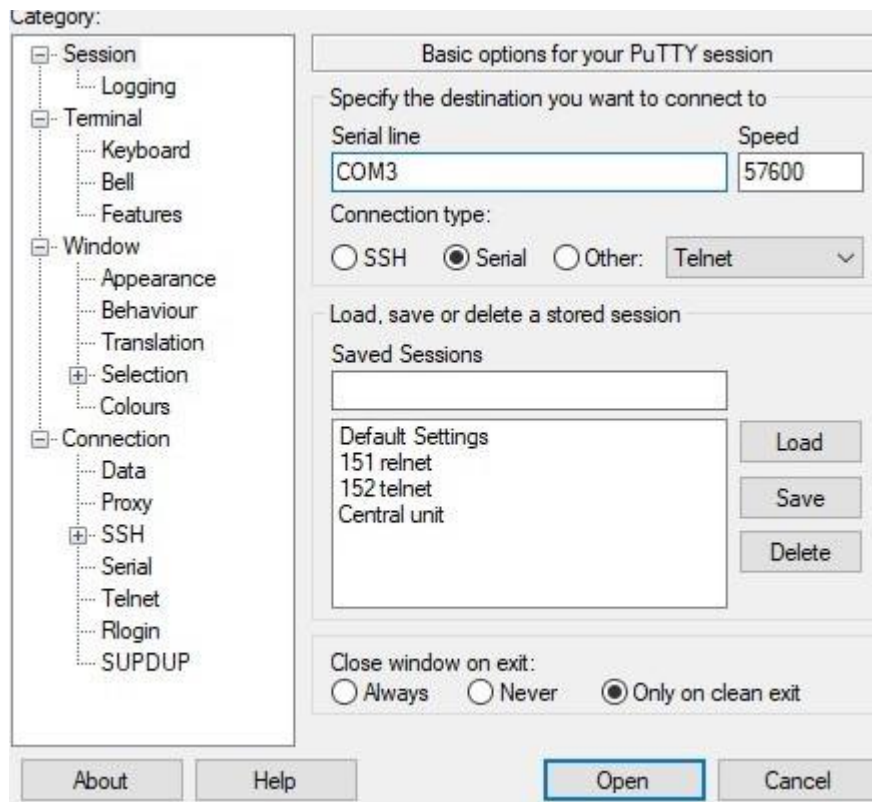


Figure 52: Opening of COM Port window

- (c) Click Open to start the session

Note: The serial window of the putty application will appear as shown in Figure 53 below:

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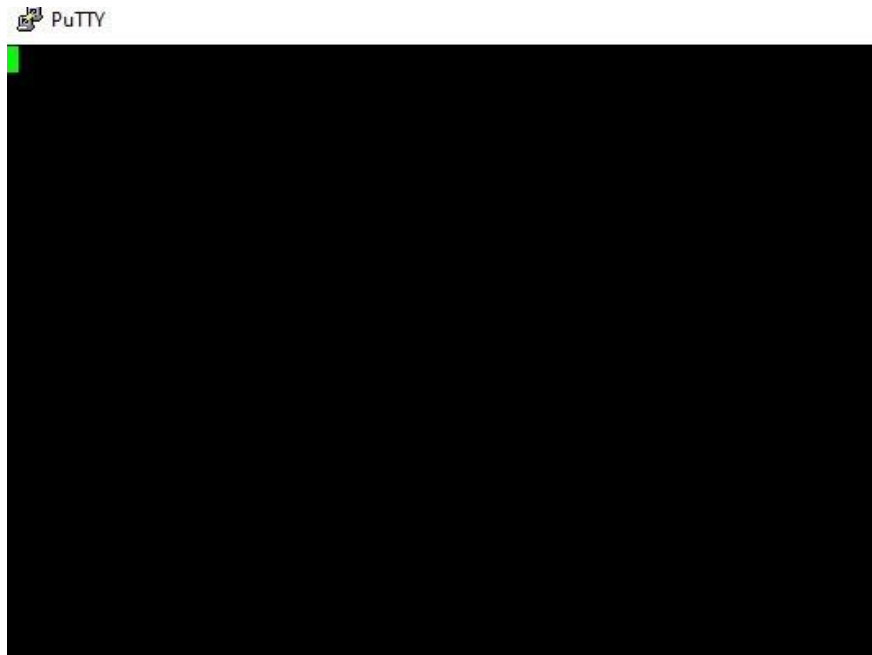
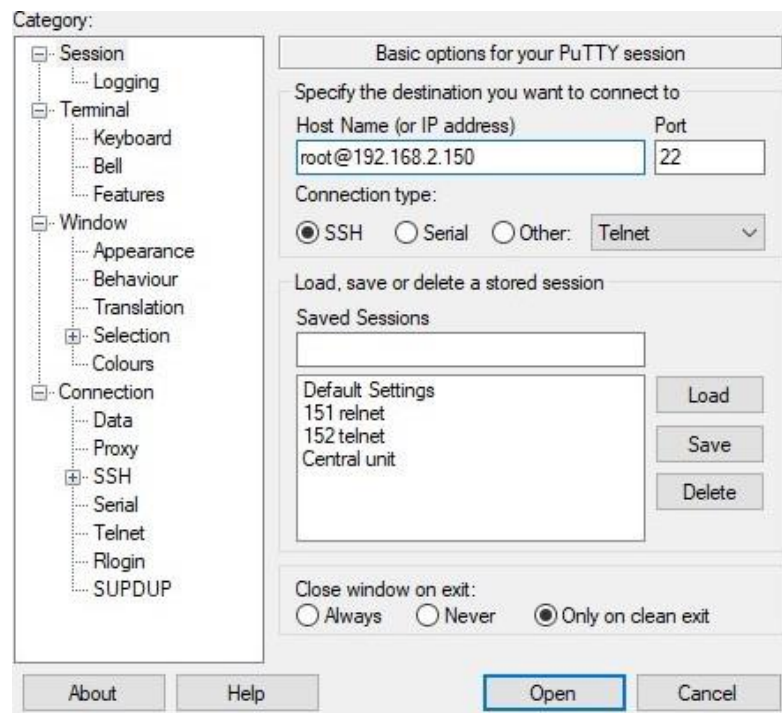


Figure 53: View of COM Port window

- 4 Connect the PC/Laptop Ethernet port with the Central Unit using a standard LAN Cable on the MGMT Port.
- 5 Open another PuTTY session by Host Name root@192.168.2.150 and the port number 22 as shown in Figure 54.



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Figure 54: Opening of Putty SSH window

- (d) Click Open to start the session
- (e) The password for logging is elcom@1234.
- (f) After logging in, apply the command in the terminal **microcom /dev/ttyAMA2 -s 57600** as shown in Figure 55 (Note: The GPS connections at CSU unit are on Utility-2 port to verify the GPS Interface), Now type the content in SSH window after the above-mentioned command.

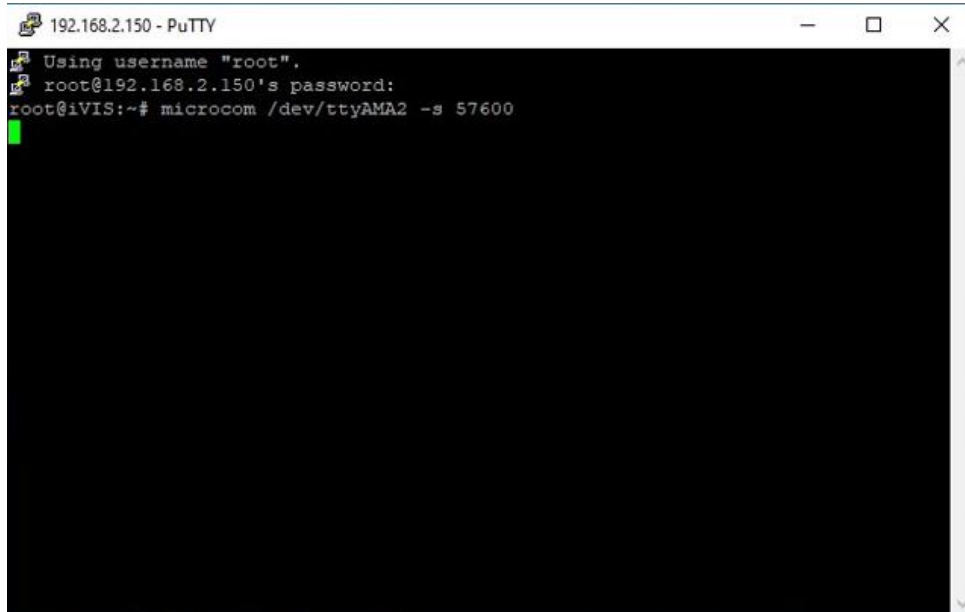


Figure 55: Logging of SSH window

- f) Verifying the characters are visible on COM3 terminal, as shown in figure below.



Figure 56: Data Receive on COM Terminal

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		Verified By	Prafulla Sarode	22.07.2026

C. Expected Result

Result: Verify the GPS data transfer: - The data/text typed on SSH terminal should be visible on the COM3 Terminal, while the content typed from the COM3 Terminal should be visible on the SSH Terminal.

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iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

12 TMR Sheet (Test Measurement Record)**12.1 Physical Parameters of Central Switch Unit**

S. No	Dimension Description	Measure and tolerance	Observed Value
1	Length	210 mm (± 1 mm)	
2	Height	197 mm (± 1 mm)	
3	Depth	86 mm (± 1 mm)	
4	Weight	2.250 Kg $\pm$ 0. 200 Kg	
5	Colour	Olive-Green	
6	Radio Connector	4 nos (R/A-1 to R/A4) Bottom side	
7	Connector for Connecting Cable	5 nos (J1 to J4) at Bottom side and (J5) on Top side	
8	SPE Connector	1 no (J6) at Top side of CSU	
9	24V DC Power Connector	1 no at Top side of CSU	
10	Power on/off switch	1 no at Top side of CSU	
11	Management Eth Connector	1 no (MGMT) at Top Side	
12	Utility-1	1 no at the Bottom of CSU	
13	Utility-2	1 no at the Top side of CSU	
14	Antenna Port	1 no at Bottom side of CSU	
15	Label present on back side of cabinet with proper serial number	YES/NO	
16	Unit is without any damage	YES/NO	
17	Paint is proper and there are no scratches	YES/NO	

Table 22: TMR Sheet for Central Switch Unit

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

12.2 Physical Parameters of User Units of CMTV

S. No	Dimension Description	Measure and tolerance	Observed Value on			
			User Unit – 1 (CMDR)	User Unit – 2 (IO)	User Unit – 3 (DVR)	User Unit – 4 (CO-DVR)
1	Length	147 ± 1 mm				
2	Length with mounting cuts	180 ± 1 mm				
3	Height without protective hinges	142 ± 1 mm				
4	Height with protective hinges	157 ± 1 mm				
5	Depth	84 ± 1 mm				
6	Weight	1.500 ± 0.100 Kg				
7	Colour	Olive-Green				
8	Headset Connector	1 no (Headset)				
9	ETH & Power Connector	1 no (ETH)				
10	SPE & Power Connector	1 no (SPE)				
11	QWERTY Keyboard Connector	1 no (KBRD)				
12	Input Power Connector	0 no (PWR)				
13	Volume Knob	1 no				
14	Volume Knob Position	11 positions				
15	Mode Knob	1 no				
16	Mode Knob position	6 positions				
17	OLED Display	1 no				
18	Built-In Keypad Buttons	8 no's				

Table 23: TMR Sheet for User Unit in CMTV

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

Physical Parameters of User Units of AATV

S. No	Dimension Description	Measure and tolerance	Observed Value on		
			User Unit – 1 (DOC)	User Unit – 2 (DVR)	User Unit – 3 (CO-DVR)
1	Length	147 ± 1 mm			
2	Length with mounting cuts	180 ± 1 mm			
3	Height without protective hinges	142 ± 1 mm			
4	Height with protective hinges	157 ± 1 mm			
5	Depth	84 ± 1 mm			
6	Weight	1.500 ± 0.100 Kg			
7	Colour	Olive-Green			
8	Headset Connector	1 no (Headset)			
9	ETH & Power Connector	1 no (ETH)			
10	SPE & Power Connector	1 no (SPE)			
11	QWERTY Keyboard Connector	1 no (KBRD)			
12	Input Power Connector	0 no (PWR)			
13	Volume Knob	1 no			
14	Volume Knob Position	11 positions			
15	Mode Knob	1 no			
16	Mode Knob position	6 positions			
17	OLED Display	1 no			
18	Built-In Keypad Buttons	8 no's			

Table 24: TMR Sheet for User Unit in AATV

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

12.3 Physical Parameters of External Keyboard

S. No	Dimension Description	Measure and tolerance	Observed Value
1	Length	136.00 ± 0.50 mm	
2	Width	76.00 ± 0.50 mm	
3	Depth	20.00 ± 0.50 mm	
4	Color	Olive-Green	
5	Keyboard Buttons	49 Nos	
6	Weight	0.270 ± 0.050 kg	

Table 25: TMR Sheet for External Keyboard

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
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12.4 Physical Parameters of Stick Radio Dock Station

S. No	Dimension Description	Measure and tolerance	Observed Value
1	Length	150.00 ± 0.50 mm	
2	Height	90.00 ± 0.50 mm	
3	Depth	85.00 ± 0.50 mm	
4	Color	Olive-Green	
5	Input Power Connector	1 Nos	
6	Weight	0.270 ± 0.050 kg	

Table 26: TMR Sheet for External Keyboard

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 ELCOM INNOVATIONS	Draft Acceptance Test Procedure	Doc ID. 1791-UCZN10331	Doc Rev. PA-11	Date
iVIS (UCZN10331)		Prepared By	Shreyash K.	22.07.2026
		Verified By	Prafulla Sarode	22.07.2026

**THE AUTHORITY OF MAKING ANY CHANGES IN THE TEST PARAMETER
RESTS ONLY WITH AHSP/ DQA (ICV)**